



High heat flux immersion cooling in data centers: Quantitative bypass control and fluid-structure compatibility across dielectric fluids and heat sink designs

Inwoo Chun, Hongseok Choi , Yongjoo Jun, Sangwook Lee, Hoseong Lee

Department of Mechanical Engineering, Korea University, 409 Innovation Hall Bldg, Anam-Dong, Seongbuk-Gu, Seoul, Republic of Korea

ARTICLE INFO

Keywords:

Immersion cooling
Data center
Dielectric
Thermal management
Metal foam

ABSTRACT

High-density artificial intelligence workloads have caused escalating heat fluxes in data centers, posing thermal risks that demand advanced thermal management and high energy efficiency. Single-phase immersion cooling systems have been extensively researched, yet existing investigations have been limited to heat fluxes below $30 \text{ W}\cdot\text{cm}^{-2}$, a range that leaves its high heat flux regime and the quantitative impact of fluid-structure interactions and thermal enhancers unexplored. To bridge these voids, this study investigated single-phase immersion cooling under chip-level heat loads exceeding 700 W per chip. The open ratio is introduced as a quantitative metric for bypass control, and its effects on thermal resistance and pumping power are evaluated using FC-40 and GC-5X across plate fin and metal foam-assisted configurations. Despite superior thermophysical properties, GC-5X recorded a chip temperature 23.0 K higher than FC-40 under unconstrained flow. Open ratio reduction from 100% to 0% decreased thermal resistance by 45.6% for GC-5X, 7.5 times greater than FC-40, at the cost of a 37.8% pumping power increase. In metal foam configurations, GC-5X reached 112.3°C at open ratio 100% yet recovered to 59.3°C at open ratio 0% , while FC-40 decreased from 57.3°C to 46.7°C . The metal foam configuration at open ratio 0% demonstrated the most stable thermal performance, with FC-40 below 79°C up to $70 \text{ W}\cdot\text{cm}^{-2}$ while GC-5X approached this threshold at $50 \text{ W}\cdot\text{cm}^{-2}$. Ultimately, these findings demonstrate that thermal performance is governed by the interplay between fluid properties and flow control, providing a quantitative design basis for immersion cooling systems under next-generation high-power conditions.

1. Introduction

Data centers have experienced a sustained increase in computational demand driven by the rapid proliferation of cloud computing, big data analytics, and artificial intelligence (AI) and machine learning (ML) workloads [1]. These intensified workloads have led to a significant rise in thermal design power (TDP) and heat flux [2]. In practice, high-end hardware like the NVIDIA H100 frequently surpasses the 700 W threshold, far exceeding the 300 W load of contemporary processors [3]. As these extreme thermal loads directly threaten hardware stability and system reliability, efficient cooling is now critical for the performance of next-generation infrastructures. Beyond managing high-density thermal loads, energy efficiency is a paramount concern as cooling accounts for $30\text{--}50\%$ of total data center electricity usage [4]. With the sector's global consumption projected to reach 1000 TWh by 2030, optimizing thermal management has become an urgent mandate [5]. Therefore, there is a

critical need for innovative cooling architectures that can effectively dissipate extreme heat loads while simultaneously achieving a lower power usage effectiveness (PUE) to ensure infrastructure sustainability.

To meet these rigorous requirements, the industry is shifting away from traditional air cooling, which is increasingly constrained by the low thermal conductivity and specific heat capacity of air [6,7]. While indirect liquid cooling via cold plates provides an improvement, it necessitates complex piping and poses leakage risks while failing to address secondary heat-generating components like voltage regulator modules (VRM) and memory [8–11]. As a superior alternative, immersion cooling involves submerging entire servers in non-conductive dielectric fluids [12,13]. By establishing direct thermal contact with all internal surfaces, this method eliminates energy-intensive cooling fans and facilitates omnidirectional heat transfer [14]. Single-phase immersion cooling has also recently been extended to other thermally demanding systems, such as battery thermal management, where direct dielectric fluid contact can provide effective and spatially uniform heat

* Corresponding author.

E-mail address: hslee1@korea.ac.kr (H. Lee).

<https://doi.org/10.1016/j.energy.2026.141967>

Received 19 May 2026; Received in revised form 28 June 2026; Accepted 14 July 2026

Available online 14 July 2026

0360-5442/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Nomenclature		ran	random
Symbols		s	solid
c_p	specific heat capacity [$\text{J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$]	sys	systematic
C_f	inertial coefficient	th	thermal
d	diameter [m]	total	total
d_f	fiber diameter [m]	bypass	bypass
d_p	pore diameter [m]	Abbreviations	
$\vec{g}$	gravitational vector [$\text{m}\cdot\text{s}^{-2}$]	3D	three-dimensional
K	permeability [m^2]	AI	artificial intelligence
m	mass [kg]	AC	alternating current
$\dot{m}$	mass flow rate [$\text{kg}\cdot\text{s}^{-1}$]	CFD	computational fluid dynamics
∇P	static pressure [$\text{Pa}\cdot\text{m}^{-1}$]	CPU	central processing unit
PPI	pores per inch	DAQ	data acquisition
R_{th}	thermal resistance [$\text{K}\cdot\text{W}^{-1}$]	FVM	finite volume method
Re	Reynolds number	FOM	figure of merit
S_i	source term	F-S	full scale
T	temperature [$^{\circ}\text{C}$]	GPM	gallons per minute
$\vec{u}$	velocity vector [$\text{m}\cdot\text{s}^{-1}$]	GPU	graphic processing unit
V	volume [m^3]	HPC	high-performance computing
W_{pump}	pumping power [W]	IT	information technology
k	thermal conductivity [$\text{kg}\cdot\text{m}\cdot\text{s}^{-3}\cdot\text{K}^{-1}$]	IHS	integrated heat spreader
ε	porosity	LPM	liter per minute
ω	pore density	LTE	local thermal equilibrium
μ	dynamic viscosity [$\text{kg}\cdot\text{m}^{-1}\cdot\text{s}^{-1}$]	LTNE	local thermal non-equilibrium
ρ	density [$\text{kg}\cdot\text{m}^{-3}$]	MAPE	mean absolute percentage error
u	uncertainty	MF	metal foam
Φ_{bypass}	bypass fraction [%]	ML	machine learning
η_{OR}	open ratio efficiency [$\text{K}\cdot\text{W}^{-1}/\text{W}$]	OR	open ratio
FOM	figure of merit [K^{-1}]	PCB	printed circuit board
Subscripts		PUE	power usage effectiveness
avg	average	RRMSE	relative root mean square error
eff	effective	TDP	thermal design power
f	fluid	TIM	thermal interface material
inlet	inlet	TTV	thermal test vehicle
open	open	VFR	volume flow rate
pump	pumping	VRM	voltage regulator module

removal [15,16]. Consequently, immersion cooling enables significantly higher rack densities and superior energy efficiency, providing the high-capacity thermal management strictly required for next-generation AI environments.

Extensive research has demonstrated the superior thermal performance of immersion cooling. High-density systems with up to 128 GPUs have maintained uniform chip temperatures around 70°C [17], while

anti-gravity flow configurations have achieved up to a 55.6% reduction in CPU temperatures [18]. Additionally, proactive flow control via micropumps or internal fans has been identified as a key strategy for enhancing both thermal efficiency and energy savings [19]. Beyond single-phase systems, two phase immersion cooling has proven effective for extreme 600 W loads, maintaining stable surface temperatures at 68.3°C [20,21]. Comparative analyses consistently validate this

Table 1
Literature review on server immersion cooling.

Author (year)	Coolant		Heat source		Heat sink design	Research approach			Ref.
	FC-base	Oil-base	Footprint area (cm^2)	Maximum heat flux ($\text{W}\cdot\text{cm}^{-2}$)		Bypass control	Coolant comparison	Metal foam	
Cheng, Chang and Li et al. (2020)	✓	-	16.0	4.38	Plate fin	-	-	-	[25]
Gandhi et al. (2019)	-	✓	19.12	4.97	Channel	-	-	-	[26]
Ding et al. (2024)	✓	-	52.5	4.76	Plate fin	-	-	-	[27]
Li et al. (2023)	-	✓	14.0	8.57	Pin fin/Channel	✓	-	-	[28]
Liu et al. (2024)	-	✓	14.0	10.71	Pin fin	-	-	-	[29]
Ge et al. (2024)	✓	-	9.0	18.00	Plate fin	✓	-	-	[30]
Muneeshwaran et al. (2023)	✓	-	32.6	18.38	Plate fin	✓	-	-	[31]
Shrigondekar et al. (2023)	✓	✓	32.6	18.38	Plate fin	✓	✓	-	[32]
Current study	✓	✓	32.5	30.76	Plate fin/Metal foam	✓	✓	✓	

technology's superiority, showing better temperature management than air or hybrid systems and reducing PUE by up to 20.8% [22,23]. Consequently, recent studies have focused on evaluating diverse dielectric fluids to optimize the balance between heat transfer and total system efficiency [24].

The existing literature on server immersion cooling has been comparatively reviewed, as summarized in Table 1. Cheng et al. presented an innovative immersion cooling system using Novec 7100 dielectric fluid and plate fin heat sink, evaluating performance under coolant velocities of 0.4 to 1.2 m·s⁻¹ [25]. Their findings confirmed that higher flow speeds lowered the average temperatures from 27.05 to 26.86°C and decreased the final standard deviation of temperature variations to 0.05 K. Additionally, Gandhi et al. investigated the thermal optimization of a 1U server with 95 W TDP processors cooled via immersion cooling [26]. Their computational analysis revealed that employing EC-100 or mineral oil at a 1 GPM flow rate effectively stabilizes CPU temperatures at approximately 70°C, even when the dielectric fluid is supplied at an elevated inlet temperature of 45°C. Ding et al. demonstrated that performance plateaus at a critical flow rate of 8 LPM, with higher speeds primarily increasing flow resistance [27]. Their study introduced micropump-based turbulence and fan-based redistribution, which successfully lowered maximum CPU temperatures by 6.1% and 5.0%, respectively. Focusing on design optimization, Li et al. optimized a 1U server immersion cooling system using EC-110 coolant under flow rate of 0.1 to 0.5 m³·h⁻¹ and inlet temperature of 30 to 38°C [28]. The results showed that interactions among design and flow parameters strongly influence maximum temperature and pressure loss, and the weighting between these objectives can change the optimal configuration, leading to temperature differences and pressure loss up to 19.8°C and 18.6 Pa. In addition, Liu et al. optimized a single-phase immersion cooling tank for 20 1U servers using Ice Core 797 dielectric coolant [29]. Their results showed that adding a porous panel reduced the maximum temperature and temperature difference by 1.2°C and 3.1°C, respectively, although the pressure loss increased by 428.6 Pa. Ge et al. evaluated the performance of an immersion cooling system using FC-3283 dielectric fluid compared to traditional air-cooling for a server motherboard [30]. Their simulation results showed that the system reduced the maximum CPU temperature by 70.13% and increased the average convective heat transfer coefficient by 287.5% through the addition of baffles to optimize flow distribution. Muneeshwaran et al. also evaluated the performance of a single-phase immersion cooling system using FC-40 dielectric fluid, considering variables such as flow rate, inlet-outlet configurations, heat load, and coolant inlet temperature [31]. Their results indicated that at a maximum flow rate of 3 LPM and a heat load of 600 W, the T-type configuration reduced thermal resistance by 12.6% compared to the Z-type. Similarly, Shrigondekar et al. compared FC-40 and PAO-6 in a 1U single-phase immersion cooling system under heat loads of 200–600 W, coolant flow rates of 1–3 LPM, inlet temperatures of 15–35°C, bypass gaps of 0–5 mm, and fin pitches of 1.2 and 2.2 mm [32]. The study found that a 2.2 mm fin pitch heat sink performs better than a 1.2 mm with PAO-6 due to its higher viscosity causing greater flow resistance and thermal resistance in narrower channels, opposite to the trend observed with FC-40. For a consistent comparison among these studies, the heat flux values in Table 1 were calculated by dividing the maximum applied heat load by the heat-source footprint area, which was obtained from the reported dimensions of the CPU, heater, or TTV (Thermal Test Vehicle).

Beyond these established single-phase immersion cooling strategies, various structural enhancers have been increasingly explored to further maximize thermal performance. In broader engineering fields outside of immersion cooling, metal foams have seen extensive adoption as high-efficiency thermal solutions [33–36]. By promoting structural turbulence and providing significantly extended surface areas, these porous structures effectively disrupt thermal boundary layer development to maximize convective heat transfer. Consequently, given their proven efficacy in diverse sectors, metal foams offer a promising pathway for

integration into immersion cooling systems to address the escalating heat densities of next-generation processors.

Throughout the literature review, numerous studies have significantly advanced the thermal performance of immersion cooling across diverse system configurations. Nevertheless, a critical disconnect remains between academic findings and the demands of next-generation high-power computing. As summarized in Table 1, within the context of single-phase immersion cooling systems, existing investigations have been predominantly limited to heat fluxes below 30 W·cm⁻², leaving the thermal behavior of dielectric fluids and heat sink structures under kilowatt-scale chip-level power densities entirely uncharacterized. Furthermore, while bypass suppression has been identified as an effective thermal management strategy, its quantitative impact remains unestablished under the extreme heat flux conditions where such control is critically needed. Specifically, how fluid-specific thermophysical characteristics govern structural sensitivity to bypass across different architectures has not been systematically quantified. Finally, although metal foams offer strong potential as thermal enhancers, their viability under high heat densities and their compatibility with dielectric fluids of contrasting thermophysical profiles remain largely unaddressed.

To address these limitations, this study presents the systematic investigation of immersion cooling under chip-level heat loads exceeding 700 W per chip, establishing a fluid-structure compatibility framework that couples dielectric fluid thermophysical properties with structural flow control. To quantify the role of bypass suppression under such conditions, the open ratio (OR) is introduced as the first quantitative metric for bypass control in immersion cooling. Through this metric, this study demonstrates how fluid-specific thermophysical characteristics govern structural flow sensitivity and drive a performance inversion between fluids under full confinement. Building on this framework, metal foam heat sinks are numerically investigated to reveal how porous media environments amplify bypass sensitivity and how fluid thermophysical properties determine the effectiveness of foam integration and optimal foam architecture. These investigations collectively identify which structural and fluid conditions sustain stable thermal management under extreme heat densities, offering a practical design direction for next-generation high-power immersion cooling systems.

To achieve these objectives, a high-fidelity numerical model was developed and validated against experimental data from both 1U and 3U server configurations. The validated model was applied to systematically investigate FC-40 and GC-5X across plate fin and metal foam-assisted heat sink designs under varying OR conditions. By visualizing flow patterns and quantifying thermal resistance and pumping power, this study identifies the physical mechanisms by which fluid thermophysical properties and structural design collectively govern cooling efficiency under high heat flux environments.

2. Research approach

2.1. Experiment

To evaluate the thermal performance of the immersion cooling system, an experimental setup was developed to simulate a server environment under high-heat-flux conditions. In actual data centers, the standard server rack unit (1U) is defined by a height of 1.75 inches (44.45 mm). This study strictly adheres to this industry standard to ensure the experimental immersion tank replicates the physical environment of commercial server deployments. While the 3U configuration was employed to evaluate cooling performance in a high-density, multi-board environment, additional experiments were conducted at the 1U scale. The 1U setup allows for a precise evaluation of cooling enhancement achieved through flow concentration within a single server unit. Specifically, the 3U configuration employed three printed circuit boards (PCB) equipped with two heat sources each, whereas the 1U setup used a single board. Furthermore, a flow control block was implemented in the

1U configuration to investigate the impact of forced flow concentration on thermal management efficiency.

The comprehensive configuration of the cooling cycle is illustrated in Fig. 1 (a). The system consists of two distinct cooling loops thermally coupled via a plate heat exchanger: a primary dielectric fluid loop for server cooling and a secondary chiller loop for heat rejection. The primary loop facilitates the circulation of the dielectric fluid through the immersion tank, which is constructed from SUS304 stainless steel. The tank is designed with a single inlet at the bottom and two outlets on the

upper sides to ensure an upward fluid flow that effectively cools the submerged heat sources. As shown in Fig. 1 (b), the front of the tank features a tempered glass window, allowing for real-time observation of the fluid flow and any thermal phenomena during operation. To minimize unintended heat loss through the stainless-steel walls, the entire tank was wrapped with a double layer of high-density thermal insulation.

A magnetic pump (PM-300PIS, Wilo pump) circulates the dielectric fluid, while a mass flow meter (KMS-2000, Daejin Instrument Co) at the

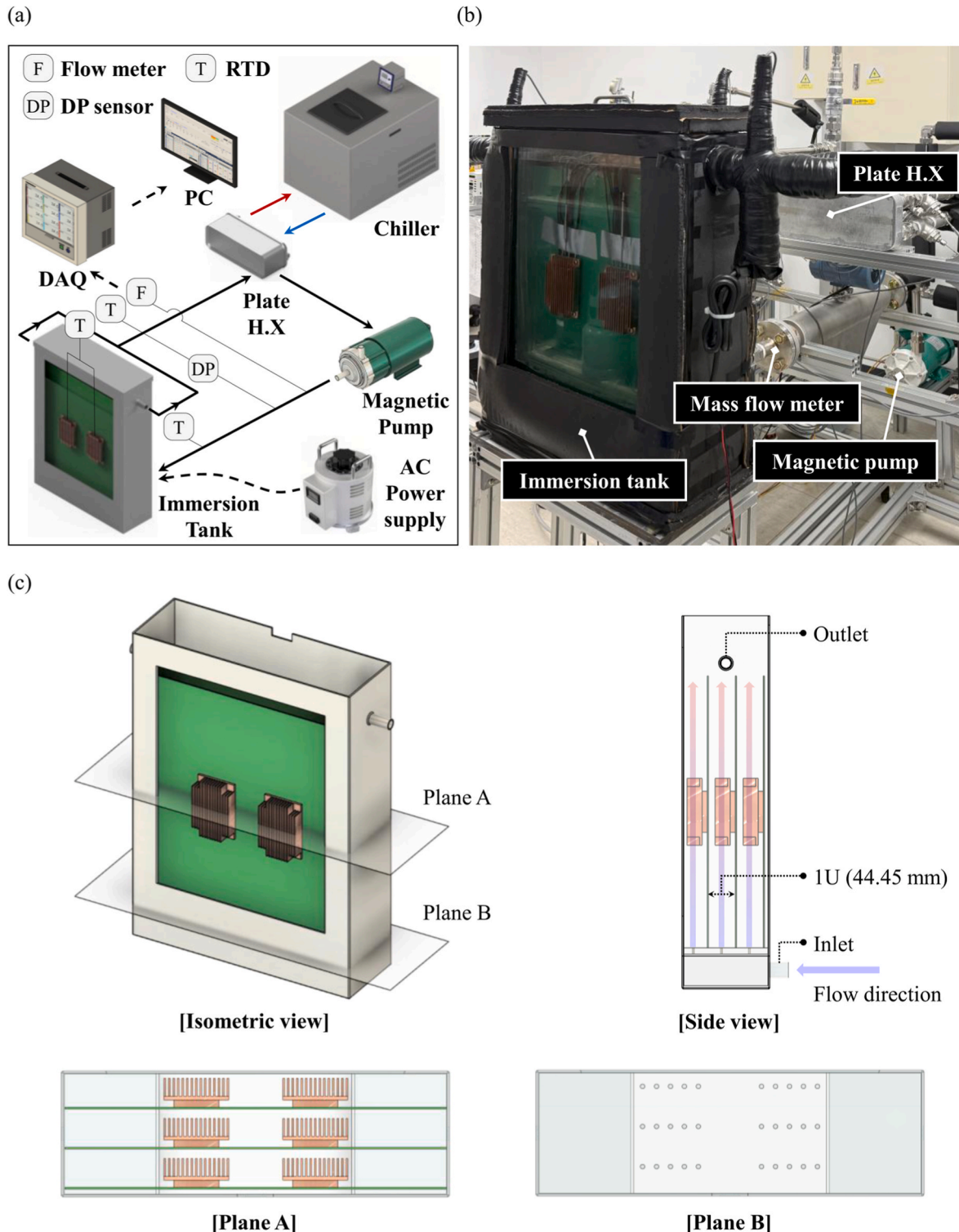


Fig. 1. (a) The schematic diagram, (b) photograph of experimental setup and (c) specification of immersion tank.

inlet precisely monitors the flow rate with an accuracy of $\pm 0.2\%$ R.D. To regulate the temperature of the incoming fluid, it exchanges heat with the secondary loop within a plate heat exchanger. The secondary loop utilizes a 50:50 mixture of water and ethylene glycol, maintained at a constant temperature by an external chiller (RXCU75C-CPTH, REXTNI). To monitor thermal behavior, RTD sensors (PT100, OMEGA Engineering) with an accuracy of $\pm 0.15^\circ\text{C}$ and an operating range of -50°C to 250°C were installed at the tank inlet and both outlets. The pressure drop across the system was measured using a differential pressure sensor with an accuracy of $\pm 0.055\%$ F.S installed between the inlet and outlet of the tank. All physical data are recorded by a DAQ system (GP20, YOKOGAWA) connected to a PC. The heating elements are powered by an external AC power supply (SD-5000, DASUNG ELECTRONICS), providing a variable voltage from 0 to 220 V to achieve the target thermal load.

The internal configuration and geometric details of the immersion tank are presented in Fig. 1 (c). The tank was designed to accommodate up to three vertical PCBs, enabling both 1U and 3U experimental configurations. As shown in the side view, Plane A and Plane B, the coolant enters from the bottom inlet, passes through the flow distribution plate, and flows upward toward the PCB and heat sink region before leaving through the outlet. Plane A indicates the relative placement of the PCBs and heat sinks, while Plane B shows the flow distribution plate. The internal slots were arranged at a standard 1U spacing of 44.45 mm, allowing the 3U configuration to be assembled by installing three PCBs at 1U intervals.

To ensure uniform and targeted delivery of the dielectric fluid from the single bottom inlet to the heat sources, a flow distribution plate was

positioned at the lower section of the tank, as shown in the side view and Plane B. This distribution plate, fabricated from transparent acrylic, was designed to rectify the incoming flow and direct it toward the mock CPUs and heat sinks. The number and diameter of the flow-distribution holes were selected according to the different roles of the 3U and 1U configurations, rather than for a direct hole-diameter performance comparison. For the 3U configuration shown in Fig. 1 (c), a plate with 30 holes of 5 mm diameter was employed because multiple PCBs and heat sources require coolant to be distributed over a relatively wide region. This design choice emphasizes flow equalization among several heat-source locations. Previous studies on flow equalization plates in single-phase immersion cooling have reported that smaller holes can improve flow distribution uniformity by increasing local flow resistance, whereas larger holes reduce pressure loss but make the flow distribution more sensitive to inlet flow rate and local nonuniformity [37]. Therefore, the 5 mm holes in the 3U configuration were selected to support more uniform coolant delivery toward multiple boards and heat sources. In contrast, the 1U configuration was intended to evaluate flow concentration and open-ratio control around a single active board, rather than coolant distribution among multiple boards. Since the number of active heat sources and flow paths was smaller than in the 3U configuration, only the central 10 holes corresponding to the single active board were used, and their diameter was increased to 10 mm to provide sufficient coolant supply to the active heat sink region while avoiding excessive flow resistance.

As depicted in Fig. 2 (a), the mock CPU assembly consisted of cartridge heaters, a copper block, indium foil, a plate fin heat sink, and a printed circuit board (PCB). The copper block was fabricated to

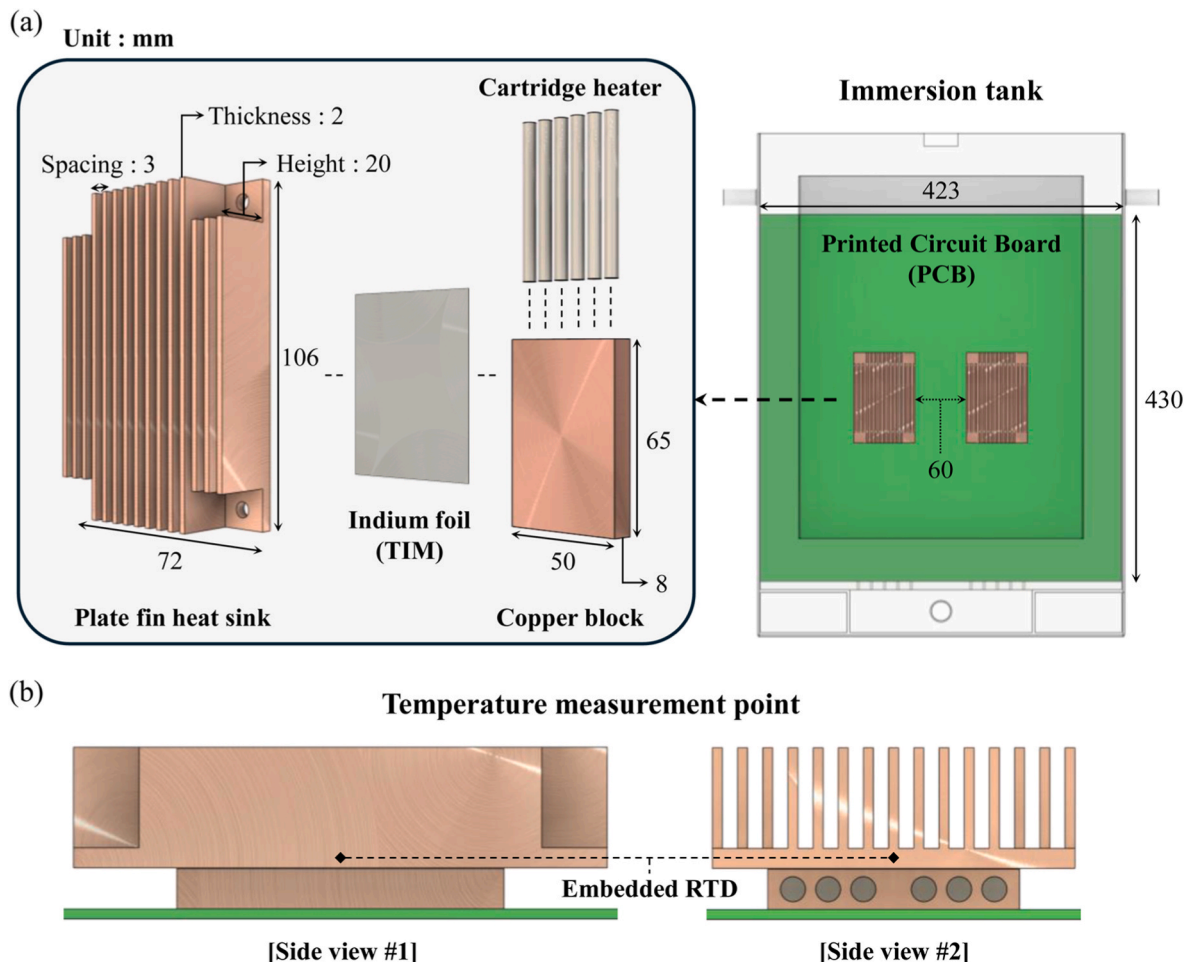


Fig. 2. Configuration of mock CPU, heat sink and printed circuit board (PCB).

represent the geometric scale of the central processing unit (CPU) and integrated heat spreader (IHS) typically used in commercial computing servers. SUS304 cartridge heaters were inserted into the copper block to generate the heat load, and each cartridge heater could provide a heat input of up to 120 W. The total heat input of each mock CPU was designed to reach a maximum of 700 W to reproduce the severe thermal environment of next-generation high-performance computing processors. The mock CPU assemblies were mounted on the PCB inside the immersion tank, and the relative placement of the two heat sources is also shown in Fig. 2 (a). The plate fin heat sink was mounted on the copper block to dissipate the generated heat, and its main geometric dimensions, including fin spacing, fin thickness, and fin height, are indicated in the figure. The copper block and heat sink were mechanically fastened using screws to maintain stable contact pressure, and indium foil with a thickness of approximately 0.1 mm was inserted at the copper-block/heat-sink interface as the thermal interface material (TIM) to reduce contact thermal resistance. The spacing between the two heat sinks was set to 60 mm to realize a representative multi-heat-source arrangement on a single PCB and to provide sufficient clearance for coolant flow and bypass formation between adjacent heat sinks. In addition, the flow control block and distribution plate were fabricated from acrylic to facilitate flow regulation.

The temperature measurement location is shown in Fig. 2 (b). An RTD sensor was embedded at the geometric center of each heat sink and used as the representative temperature monitoring point for experimental validation. Direct temperature measurement at the copper block-heat sink interface was avoided because the sensor bead and lead wire could disturb the contact interface, locally separate the heat sink from the copper block, and alter the thermal contact resistance. Therefore, the embedded RTD arrangement was selected to minimize disturbance to the copper block-heat sink interface while maintaining a repeatable thermal contact condition during assembly. The measured temperature was used as a representative heat-sink center temperature at the specified RTD location. For model validation, the CFD temperature was obtained from the same location as the embedded RTD sensor. Therefore, the experimental and numerical temperatures were compared at the same measurement point, which reduces ambiguity caused by temperature nonuniformity within the assembly.

The detailed thermophysical properties of all materials used in this experimental setup are summarized in Table 2.

To compare the cooling effectiveness based on fluid properties, two types of dielectric fluids were employed: FC-40 as a fluorocarbon-based fluid and GC-5X as an oil-based fluid. The thermophysical properties of these fluids are summarized in Table 3. Although GC-5X possesses superior specific heat and thermal conductivity, FC-40 offers advantages in flow development around the heat sources due to its lower viscosity, which is consistent with findings in previous thermal management studies.

The measurement uncertainty was evaluated to confirm the reliability of the experimental data used for CFD validation. For each directly measured variable, the systematic and random errors were combined using the root-sum-square method as

$$u_{X_i} = \sqrt{u_{X_{i,sys}}^2 + u_{X_{i,ran}}^2} \quad (1)$$

Table 2
Properties of solid materials.

Material	Density (kg·m ⁻³)	Thermal conductivity (W·m ⁻¹ ·K ⁻¹)	Specific heat (J·kg ⁻¹ ·K ⁻¹)	Ref.
Copper	8978	387.6	381	[22]
PCB	3600	42.0 (In-plane) 0.5 (Cross-plane)	720	[22]
TIM	7310	81.8	234	[38]
SUS304	8000	16.4	500	[39]
Acrylic	1180	0.21	1460	[40]

where u_{X_i} is the total error of the measured variable X_i , and $u_{X_{i,sys}}$ and $u_{X_{i,ran}}$ are the systematic and random errors, respectively. The systematic error was determined from the accuracy of each measurement device, while the random error was estimated from the standard deviation of repeated steady-state measurements. For derived quantities, the uncertainty was propagated from the contributing measured variables as

$$u_R = \sqrt{\sum_{i=1}^n \left(\frac{\partial R}{\partial X_i} u_{X_i} \right)^2} \quad (2)$$

where R is the derived quantity and X_i is an independent measured variable. The analysis was performed under a representative steady-state operating condition of $TDP = 700$ W per chip, $T_{in} = 25^\circ\text{C}$, $\dot{m} = 2100$ kg·h⁻¹. As a result, the total errors of fluid temperature, heat source temperature, mass flow rate, and pressure drop were estimated to be 0.1516°C, 0.1745°C, 0.39%, and 1.83% of the reading, respectively.

2.2. Governing equations

To simplify the simulation while maintaining physical accuracy, the following assumptions were made:

The flow is assumed to be steady-state and incompressible.

Gravitational acceleration is considered in the negative y-direction to account for buoyancy.

The metal foam is modeled as a porous medium using the Darcy-Forchheimer law.

The flow transitions between laminar and turbulent regimes in accordance with the inlet flow rate; notably, the k-epsilon turbulence model is implemented for the metal foam configurations to accurately resolve the internal flow fluctuations.

Temperature-dependent functions were applied to all FC-40 and GC-5X properties, whereas constant values were utilized for the other properties of all solid materials.

The heat sources generate uniform volumetric heat in all directions.

The following governing equations for mass, momentum, and energy conservation are employed to analyze the fluid flow and thermal characteristics. For the entire computational domain, excluding the porous media zone, the conservation laws are defined as follows:

Continuity equation:

$$\nabla \cdot (\rho_f \vec{u}_f) = 0 \quad (3)$$

Momentum equation:

$$\nabla \cdot (\rho_f \vec{u}_f) \vec{u}_f = \nabla \cdot (\mu_f \nabla \vec{u}_f) + \rho_f \vec{g} - \nabla P \quad (4)$$

Energy equation:

$$\nabla \cdot (\rho_f c_{p,f} T_f \vec{u}_f) = \nabla \cdot (k_f \nabla T_f) \quad (5)$$

$$\nabla \cdot (k_s \nabla T_s) = 0 \quad (6)$$

To evaluate the thermal performance of the heat sink with metal foam, a porous media approach was employed. In CFD simulations, the porous media model is widely utilized across various engineering applications, such as heat exchangers, filters, and catalytic converters, primarily because it allows for the simplification of complex geometries into a distributed resistance zone. This methodology offers advantages in terms of computational efficiency, as it effectively reduces the total mesh count and minimizes the computational cost compared to

Table 3
Properties of working fluid.

Material	Density (kg·m ⁻³)	Thermal conductivity (W·m ⁻¹ ·K ⁻¹)	Specific heat (J·kg ⁻¹ ·K ⁻¹)	Viscosity (kg·m ⁻¹ ·s ⁻¹)	Ref.
FC-40	2499 – 2.16 T	0.086 – 6.90·10 ⁻⁵ T	590 + 1.55 T	0.0429 – 0.162·10 ⁻³ T + 1.08·10 ⁻⁷ ·T ²	[22]
GC-5X	1049.862 – 0.75 T	0.199367 – 1.8·10 ⁻⁴ T	2076.055 + 0.3 T	0.6449 – 4.1234·10 ⁻³ T + 8.9697·10 ⁻⁶ ·T ² – 6.5833·10 ⁻⁹ ·T ³	[41]

resolving every individual ligament of the metal foam structure. Furthermore, this approach is particularly robust for capturing macro-scale flow characteristics and pressure drop patterns without compromising the physical accuracy of the simulation.

The porous-medium formulation for the metal foam was implemented using morphology-based correlations that have been adopted in previous numerical and experimental studies on metal-foam heat sinks [33,42]. In these correlations, the prescribed pore density and porosity are used to determine the representative pore-scale geometry of the foam. Specifically, the pore diameter (d_p) and fiber ligament diameter (d_f) are first calculated to describe the characteristic pore opening and ligament scale of the open-cell metal foam. Based on these geometric parameters, the permeability (K) and inertial coefficient (C_f) are then evaluated and used as the macroscopic hydraulic resistance parameters of the porous region. In the present CFD model, these calculated K and C_f values were assigned to the metal foam zone and incorporated into the Darcy–Forchheimer source term. Therefore, the effect of the metal foam morphology on coolant penetration and pressure drop was reflected through the derived resistance coefficients used in the governing equations.

For the numerical analysis of the metal foam, the conservation of mass, momentum, and energy is modeled using the Local Thermal Equilibrium (LTE) approach. Although some studies suggest the Local Thermal Non-Equilibrium (LTNE) model for high-conductivity media, the LTE assumption is adopted here due to the extremely small pore sizes and high specific surface area of the metal foams used, which ensure sufficient thermal coupling between the fluid and solid phases [43]. This approach provides a balance between physical accuracy and computational efficiency, defined by the following equations:

Continuity equation:

$$\nabla \cdot (\rho_f \vec{u}_f) = 0 \quad (7)$$

Momentum equation:

$$\frac{\nabla \cdot (\rho_f \vec{u}_f) \vec{u}_f}{\varepsilon^2} = \frac{\nabla \cdot (\mu \nabla \vec{u}_f)}{\varepsilon} + \varepsilon \rho_f \vec{g} - \nabla(\varepsilon P) - S_i \quad (8)$$

The source term represents the pressure gradient induced by the metal foam, combining the linear Darcy viscous resistance and the non-linear Forchheimer inertial resistance:

$$S_i = \frac{\mu}{K} \vec{u}_f + \frac{\rho_f C_f}{\sqrt{K}} \left| \vec{u}_f \right| \vec{u}_f \quad (9)$$

In Eq. (9), K and C_f are the permeability and inertial coefficient of the metal foam, respectively. These resistance coefficients were calculated from the foam morphology using the following correlations [33]:

$$K = 0.00073(1 - \varepsilon)^{-0.224} \left(\frac{d_f}{d_p} \right)^{-1.11} d_p^2 \quad (10)$$

$$C_f = 0.00212(1 - \varepsilon)^{-0.132} \left(\frac{d_f}{d_p} \right)^{-1.63} \quad (11)$$

The required pore-scale dimensions, d_p and d_f , were obtained from

the pore density and porosity as follows [42]:

$$d_p = \frac{0.0224}{\omega} \quad (12)$$

$$d_f = 1.18 \sqrt{\frac{1 - \varepsilon}{3\pi}} d_p \left[1 - \exp \left(-\frac{1 - \varepsilon}{0.04} \right) \right]^{-1} \quad (13)$$

Consequently, the energy conservation for both phases is integrated into a single governing equation:

$$\nabla \cdot (\varepsilon \rho_f c_{p,f} \vec{u}_f T) = \nabla \cdot (k_{\text{eff}} \nabla T) \quad (14)$$

$$k_{\text{eff}} = \varepsilon k_f + (1 - \varepsilon) k_s \quad (15)$$

The above closure relations were adopted from experimentally supported porous-medium formulations for open-cell metal foams. In this framework, the foam morphology is converted into permeability and inertial resistance coefficients, which are then imposed in the Darcy–Forchheimer source term to reproduce the macroscopic pressure-loss behavior of the metal foam. Therefore, although the metal foam assisted heat sink was not directly tested in the present experimental setup, the CFD model was constructed using established resistance correlations rather than arbitrary porous inputs. Accordingly, the metal foam results are used to evaluate the relative thermal-hydraulic trends among the investigated configurations.

2.3. Numerical model

The numerical analysis was performed in a systematic sequence involving geometric modeling, mesh generation, and solver configuration as shown in Fig. 3 (a). The 3D geometry of the immersion cooling system and the hybrid heat sink was modeled using ANSYS SpaceClaim. The design focused on the precise representation of the fluid domain and the solid components to accurately simulate the conjugate heat transfer.

The computational domain was then discretized using Fluent Meshing with a Poly-Hexcore scheme. This approach utilizes hexahedral cells in the core regions and polyhedral cells near complex boundaries, optimizing both computational speed and geometric fidelity. To capture the steep velocity and temperature gradients near the cooling surfaces, proximity-based refinement was applied to the heat sink and porous zones. To ensure the reliability of the results, a grid independence analysis was conducted by monitoring both the heat-source temperature and pressure drop. As illustrated in Fig. 3 (b), a final mesh count of 3,682,197 cells was selected, as the relative variation in the predicted temperature and pressure drop were found to be within 0.21% and 0.12%, respectively, compared to finer mesh configurations, confirming that the grid was sufficient for numerical accuracy. The mesh quality metrics and deviations in temperature and pressure drop across all tested configurations are summarized in Table 4, with the selected mesh exhibiting a minimum orthogonal quality of 0.1749 and a maximum skewness of 0.8251.

Finally, the steady-state simulations were carried out using ANSYS Fluent 2023 R1 based on the Finite Volume Method (FVM) with double-precision. For pressure-velocity coupling, the COUPLED algorithm was employed to ensure robust convergence for the forced convection system. Spatial discretization for the pressure field was handled by the PRESTO! scheme, which is essential for accurately resolving pressure

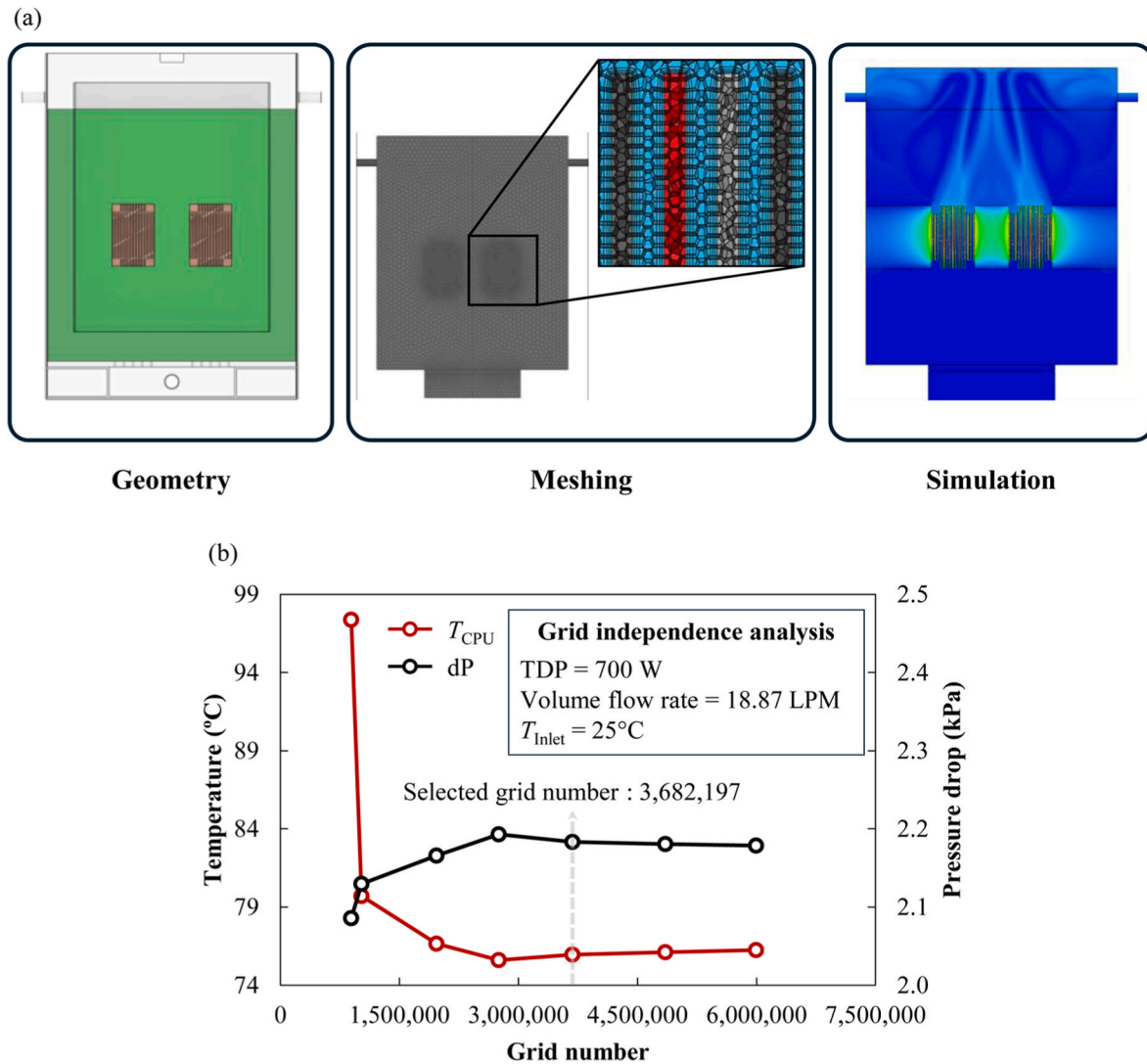


Fig. 3. (a) Procedure of numerical model and (b) result of grid independence analysis.

Table 4
Grid-independence study and mesh quality metrics.

Grid no.	Mesh elements	Min orthogonal quality	Max skewness quality	T_{CPU} deviation	dP deviation
1	887,238	0.1528	0.8472	18.15%	-2.11%
2	1,022,873	0.2000	0.7999	3.83%	-1.70%
3	1,968,012	0.1516	0.8484	1.36%	-1.25%
4	2,750,391	0.1589	0.8411	-0.45%	0.45%
5	3,682,197	0.1749	0.8251	-0.21%	0.12%
6	4,851,022	0.1505	0.8495	-0.18%	0.08%
7	5,993,204	0.1500	0.8500	-	-

distributions in porous media. The convergence was strictly monitored, with the scaled residuals required to reach below 10^{-3} for the continuity equation and 10^{-6} for the energy equation, ensuring a high level of thermal and hydraulic accuracy.

2.4. Validation

To verify the reliability of the developed numerical model, experimental measurements were conducted using the setup illustrated in Fig. 1. The experimental system was designed to evaluate the heat removal capacity and hydraulic performance of circulating dielectric

fluids. The validation encompasses two key physical quantities — heat source temperature and pressure drop — to comprehensively assess the thermal-hydraulic behavior of the immersion cooling system.

For thermal validation, FC-40 and GC-5X were tested under 1U and 3U tank operating conditions over a heat load range of 300–700 W per chip. The heat-source temperature was measured at the center of the heat sink using calibrated RTD sensors, and the CFD temperature was extracted from the same location to ensure a consistent comparison. As shown in Fig. 4 (a), the CFD results captured the experimental temperature trends with a MAPE of 2.40% and an RRMSE of 2.54%, and the maximum deviation remained within ± 3.23 K.

For hydraulic validation, the pressure-drop prediction was first evaluated at the heat sink component level and then extended to the full 1U and 3U tank configurations. This procedure was adopted because the dominant hydraulic resistance in the present immersion cooling system is generated near the heat sink, particularly within the narrow fin passages where viscous and form losses are concentrated. Therefore, the pressure drop occurring as FC-40 and GC-5X passed through the standalone heat sink was experimentally measured and compared with the CFD prediction. As shown in Fig. 4 (b), the CFD results reproduced the experimental pressure drop with a MAPE of 4.98% and an RRMSE of 6.61%, with all data points falling within a $\pm 9.55\%$ error band. These results confirm that the numerical model can reasonably capture the pressure-loss mechanism induced by the heat sink.

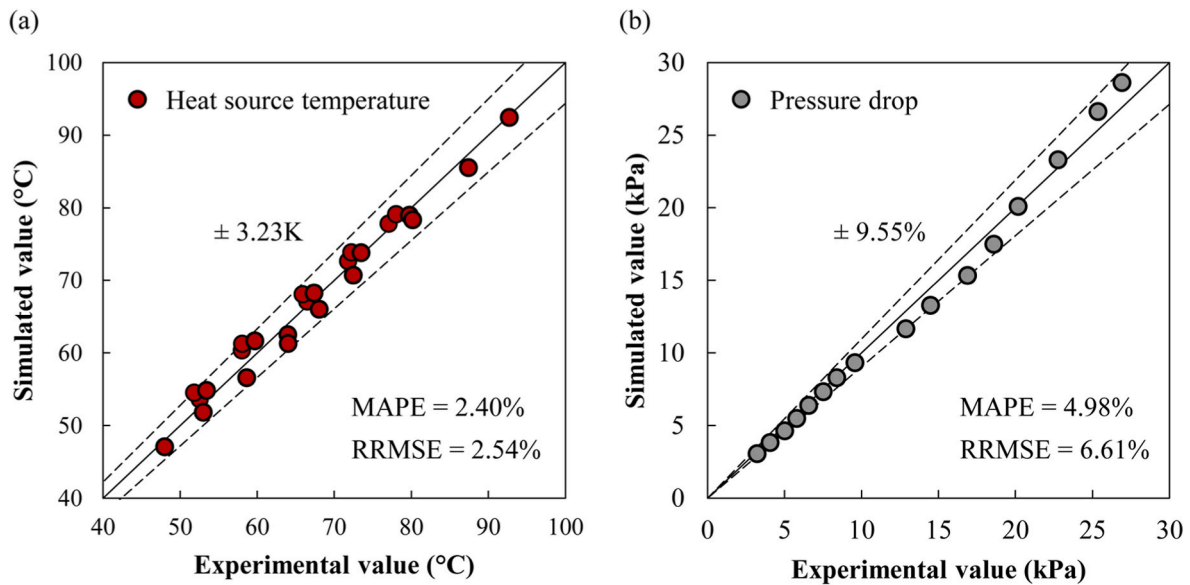


Fig. 4. Validation of server immersion cooling simulation model: (a) heat source temperature and (b) pressure drop.

Based on the validated heat-sink-level hydraulic model, the numerical domain was subsequently extended to the full immersion tank system. For the tank scale validation, the inlet-to-outlet pressure difference was measured experimentally and compared with the CFD prediction extracted at the corresponding inlet and outlet locations. The extended model predicted the tank-scale pressure drop with maximum deviations of 9.88% for FC-40 and 8.48% for GC-5X, confirming that the component-level pressure-drop validation can be reliably extended to the full tank configuration. Based on this validation, the subsequent parametric analyses were conducted using the validated CFD model.

3. Results and discussions

3.1. Thermal performance evaluation

Using the validated CFD model, the thermal and hydraulic baseline performance of two dielectric fluids, FC-40 and GC-5X, was numerically evaluated in a 3U-scale open-tank immersion cooling system. A parametric study was conducted across three key operating variables — volume flow rate, chip TDP, and coolant inlet temperature — to characterize the fundamental response of each fluid under unconstrained flow conditions. The baseline condition was set at a TDP of 700 W per chip, an inlet temperature of 25°C, and a volume flow rate of 15 LPM. It should be noted that the TDP values refer to the heat load per individual chip, with each PCB configured to host two chips, resulting in a consistent heat source distribution across the 3U tank.

As illustrated in Fig. 5 (a), the chip temperature behavior is closely tied to the convective conditions and heat flux parameters within the system. As the volume flow rate increases from 11 to 19 LPM, the chip temperature decreases due to the enhanced convective heat transfer coefficient and the thinning of the thermal boundary layer. However, this thermal improvement is notably marginal, as the chip temperature decreases from 78.3°C to 75.9°C for FC-40 and from 100.1°C to 99.1°C for GC-5X, representing a reduction of only 2.4 K and 1.0 K, respectively. This low sensitivity to flow rate reveals a significant flow bypass effect within the 3U tank, where the increased inlet flow preferentially follows low-resistance open spaces rather than penetrating the high-resistance heat sink channels. This structural limitation is further highlighted as the TDP escalates from 300 W to 700 W; the intensified heat flux necessitates a larger temperature gradient, causing an upward trend in chip temperatures. At a 700 W load, GC-5X reaches 99.6°C, while FC-40 maintains a much lower 76.6°C, reflecting FC-40's superior Reynolds

number and more effective coolant penetration into the heat sink channels. In contrast, the higher viscosity of GC-5X increases the hydraulic resistance within the narrow fin passages, causing a larger portion of the coolant to bypass the heat sink rather than actively participate in heat removal. Consequently, the favorable thermal properties of GC-5X, such as its higher specific heat and thermal conductivity, are not fully utilized under the open configuration. Similarly, rising inlet temperatures from 15°C to 35°C further elevate chip temperatures by diminishing the overall heat transfer driving force between the source and the dielectric coolant. These trends indicate that the baseline thermal performance is governed by effective coolant delivery to the heat transfer surfaces, rather than by bulk thermophysical properties alone. Notably, despite GC-5X possessing superior thermophysical properties including higher specific heat and thermal conductivity, it records a chip temperature approximately 23.0 K higher than FC-40 under identical baseline conditions.

In contrast to the thermal trends, the total pressure drop in Fig. 5 (b) is governed by inertial effects and structural resistance rather than thermal loads. The system exhibits high sensitivity to flow rate, rising sharply from 0.81 to 2.21 kPa for FC-40 and 0.97 to 2.15 kPa for GC-5X, with the steeper rise for FC-40 attributed to its higher density intensifying momentum losses. In contrast, pressure drop remains largely insensitive to both TDP and inlet temperature variations — approximately 1.43 kPa for FC-40 and 1.53 kPa for GC-5X — owing to the single-phase nature of the system where bulk temperature changes are insufficient to significantly alter fluid properties, and where form drag dominates over viscous friction within the complex heat sink geometry.

To examine the energetic implications of the observed bypass behavior, Fig. 6 presents thermal resistance and pumping power as a function of volume flow rate. Despite a substantial increase in pumping power — from 0.1489 to 0.7013 W for FC-40 and 0.1786 to 0.6916 W for GC-5X across the 11–19 LPM range — the thermal resistance of both fluids remains largely unchanged, with FC-40 decreasing marginally from 0.0761 to 0.0727 K·W⁻¹ and GC-5X remaining nearly invariant at approximately 0.1065 K·W⁻¹. This decoupling between hydraulic input and thermal output is a direct consequence of the flow bypass phenomenon: the additional fluid volume introduced by increasing the flow rate preferentially follows low-resistance open paths surrounding the heat sink rather than contributing to active heat removal at the fin surfaces. As a result, the incremental pumping expenditure yields negligible thermal return for either fluid, confirming that flow rate augmentation alone is an ineffective strategy in open-tank

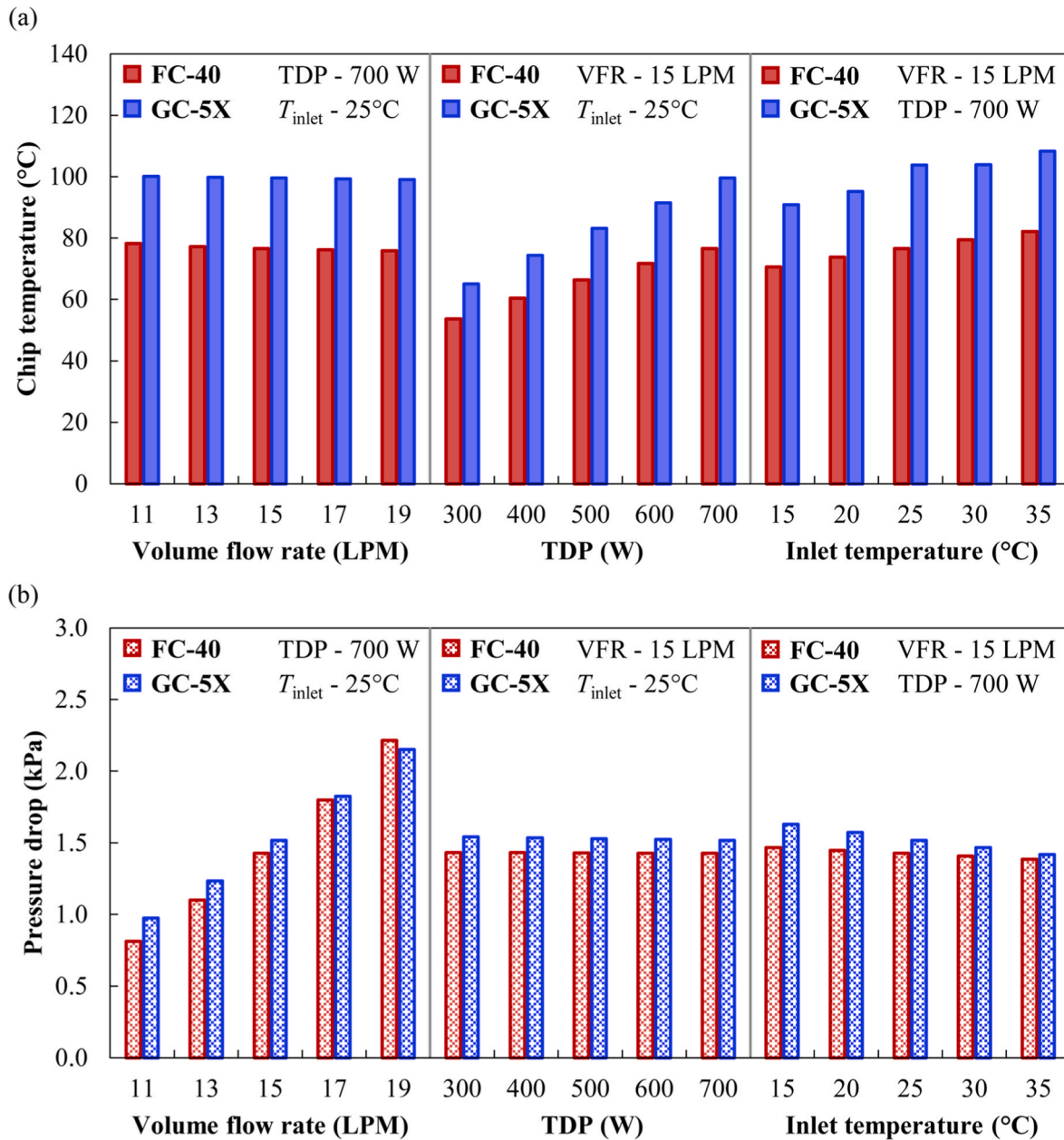


Fig. 5. Comparative analysis of (a) chip temperature and (b) pressure drop for different dielectric coolants.

configurations where bypass flow remains structurally uncontrolled.

The velocity contours in Fig. 7 provide a physical basis for the bypass-driven performance gap identified in Figs. 5 and 6. As shown in Fig. 7 (a), the dielectric fluid rising from the inlet encounters high hydraulic resistance as it approaches the heat sink, causing a portion of the flow to divert laterally or recirculate downward rather than penetrating the assembly. FC-40 exhibits a more centralized and higher-velocity flow profile compared to GC-5X, driven by its lower dynamic viscosity and higher density, while the heated fluid is efficiently discharged toward the upper outlet by both buoyancy and flow inertia.

The cross-sectional views at Planes A, B, and C in Fig. 7 (b) clearly manifest the flow bypass effect. For both fluids, a significant portion of the upward flow diverts toward the unobstructed open spaces surrounding the heat sinks, following the path of least resistance rather than penetrating the narrow fin channels. Flow velocity is noticeably higher at the lateral edges of the heat sink, while the core fin region remains relatively low. At Plane A, FC-40 demonstrates strong penetration into the fin arrays; as the fluid rises through Planes B and C,

increasing hydraulic resistance redistributes the flow toward unobstructed lateral and upper spaces. In contrast, GC-5X displays consistently suppressed velocities from Plane A onwards, with its high viscous resistance preventing effective fin entry and resulting in extensive stagnation zones by Plane C.

The superior thermal performance of FC-40, despite GC-5X possessing higher specific heat and thermal conductivity, is thus visually substantiated in Fig. 7. FC-40 maintains a more vigorous and deeper flow into the lower and central sections of the heat sink, enabling active fluid renewal near the fin surfaces that effectively thins the thermal boundary layer and yields a higher convective heat transfer coefficient. These observations confirm that augmenting the flow rate primarily exacerbates bypass flow rather than enhancing heat removal within the critical heat-generating regions, suggesting that deliberate thermal management strategies targeting flow distribution are necessary to fully realize the thermophysical potential of high-viscosity dielectric fluids.

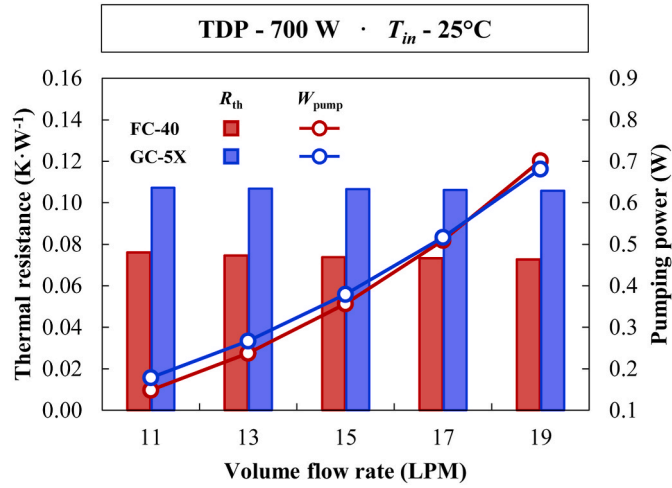


Fig. 6. Effect of volume flow rate on thermal resistance and pumping power.

3.2. Structural bypass control: open ratio

The preceding analysis established that flow bypass is the governing mechanism limiting the thermal performance of dielectric fluids in open-tank configurations. To quantitatively characterize and control this bypass phenomenon, the open ratio (OR) is introduced as a structural design variable. As illustrated in Fig. 8, the OR is defined as the volumetric fraction of the unobstructed fluid inlet space relative to the total heat sink inlet volume:

$$OR = \frac{V_{Open}}{V_{Total}} \times 100 (\%)$$

where the flow control blocks are applied symmetrically from both lateral edges, ranging from OR = 0% — where all incoming fluid is

directed through the heat source region — to OR = 100%, where no flow control plate is present and fluid bypasses freely. An intermediate case of OR = 50% was also evaluated, where half of the bypass area is blocked using a solid fin-like structure to partially constrain the flow. This parametric definition is geometry-specific and intended to quantify the degree of bypass suppression in the present 1U configuration, which differs from the 3U setup in the elimination of the top gap and the use of different hole sizes in the flow distribution plate as mentioned.

As illustrated in Fig. 9, both fluids show a gradual reduction in chip temperature and thermal resistance as the OR decreases. For FC-40, the temperature drop is relatively moderate, as its low viscosity allows for some penetration into the fin array even in the fully open state (OR 100%). In contrast, GC-5X exhibits a more pronounced sensitivity to structural control due to its strong tendency for flow bypass. Notably, at OR 0% where the bypass path is completely blocked, GC-5X achieves a lower chip temperature than FC-40, resulting in a performance inversion. This is attributed to the superior thermophysical properties of GC-5X — higher specific heat and thermal conductivity — which are effectively utilized at the heat sink surfaces under full confinement. Hydraulically, FC-40 shows low sensitivity to the reduction in OR, maintaining a stable pressure profile, whereas the pressure drop for GC-5X surges to approximately 0.94 kPa at OR 0%, driven by elevated local velocities amplifying the viscous shear resistance inherent to the high-viscosity fluid.

To quantitatively assess the structural sensitivity of each fluid, Table 5 summarizes the changes in thermal resistance and pumping power across the full OR sweep. As the OR decreases from 100% to 0%, the thermal resistance of FC-40 decreases by $0.0045 \text{ K}\cdot\text{W}^{-1}$, a reduction of 6.1%, while GC-5X shows a far more dramatic reduction of $0.0430 \text{ K}\cdot\text{W}^{-1}$, equivalent to 45.6% — approximately 7.5 times greater. In terms of hydraulic cost, the pumping power of FC-40 increases by only 0.0038 W, a rise of 4.0%, whereas GC-5X incurs a substantially larger penalty of 0.0430 W, corresponding to 37.8%.

The OR efficiency index η_{OR} was defined to quantify the thermal

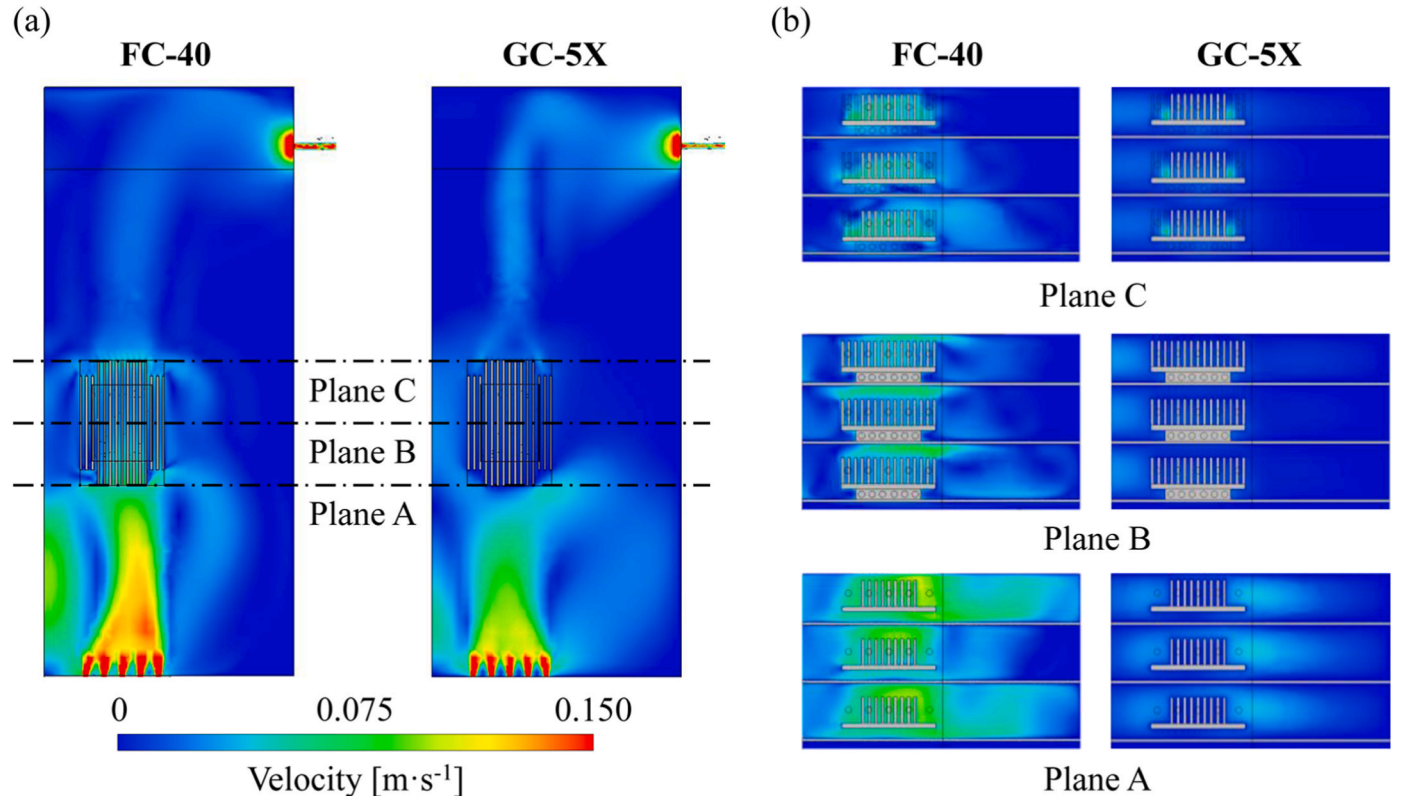


Fig. 7. Comparison of (a) front-view velocity contour and (b) local flows near the heat sink for FC-40 and GC-5X single-phase immersion cooling systems.

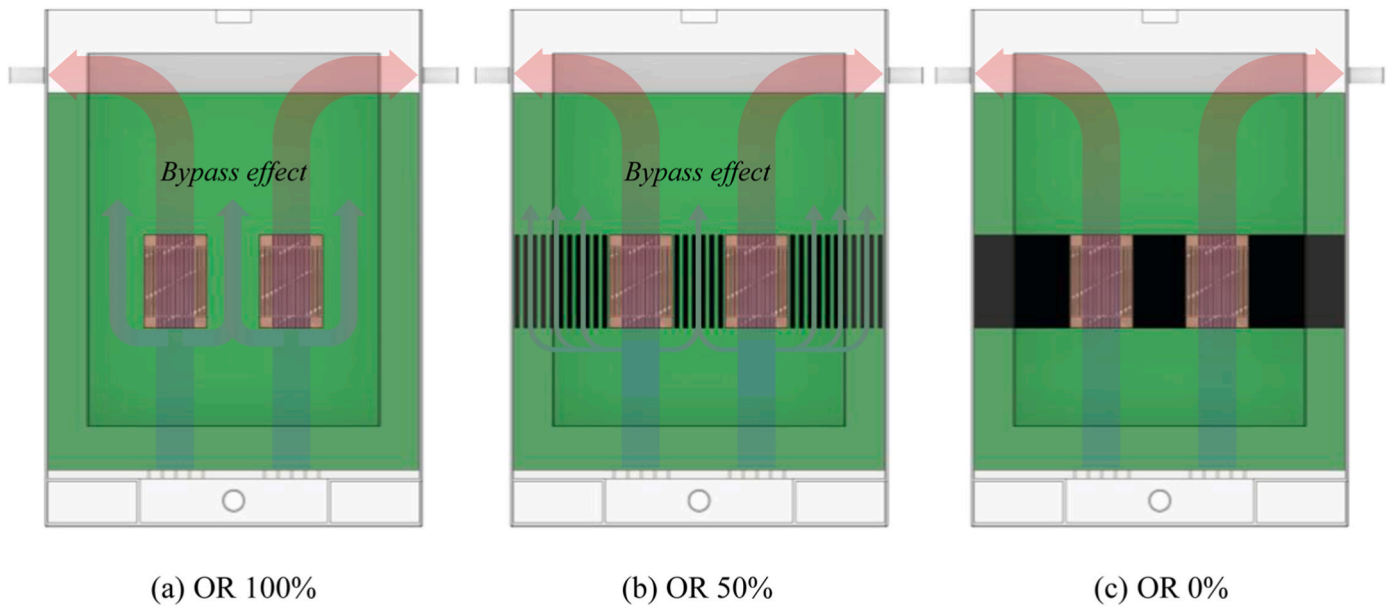


Fig. 8. Schematic of open ratio configurations: (a) OR 100%, (b) OR 50%, and (c) OR 0%.

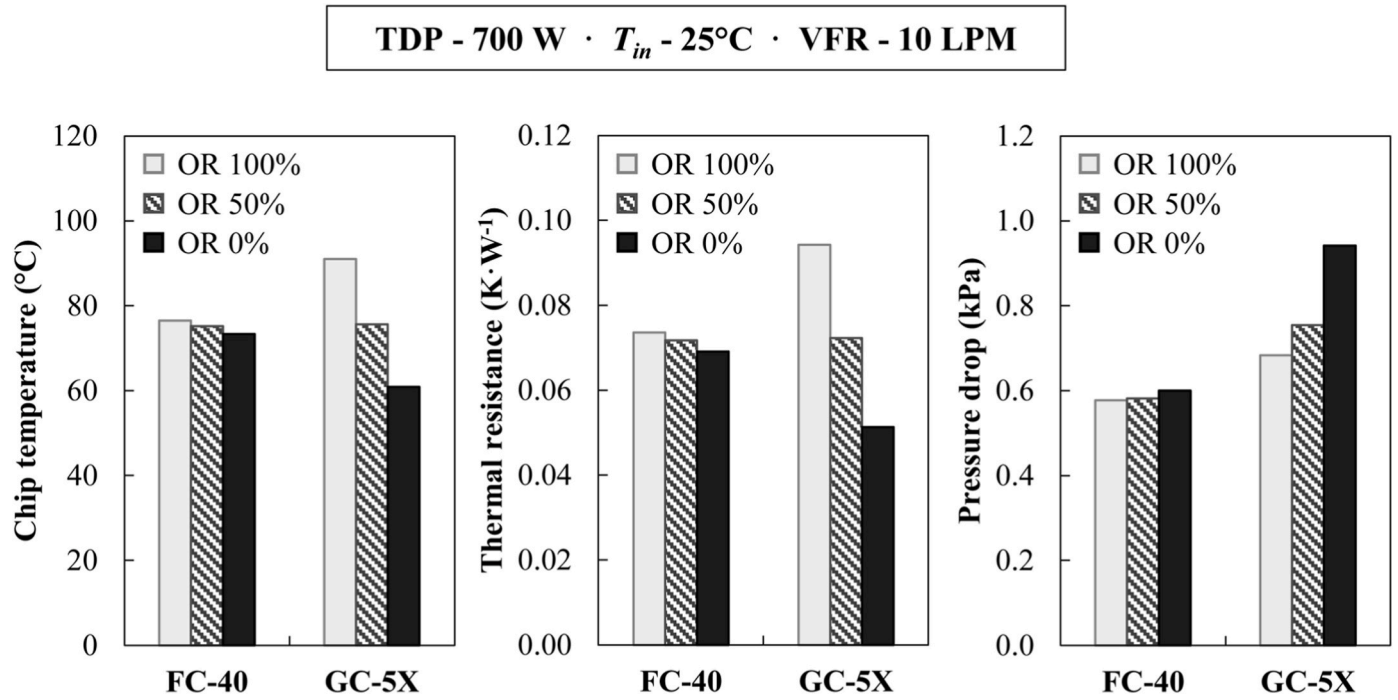


Fig. 9. Effect of open ratio on thermal performance and pressure drop for FC-40 and GC-5X.

Table 5

Thermal-hydraulic performance as a function of open ratio for FC-40 and GC-5X in the plate fin heat sink.

Parameter	R_{th} (K·W ⁻¹)		W_{pump} (W)		η_{OR} (K·W ⁻¹ /W)		FOM (K ⁻¹)	
	FC-40	GC-5X	FC-40	GC-5X	FC-40	GC-5X	FC-40	GC-5X
OR 100%	0.0736	0.0942	0.0961	0.1139	-	-	141.4	93.1
OR 50%	0.0717	0.0723	0.0969	0.1258	-	-	143.9	109.9
OR 0%	0.0691	0.0512	0.1000	0.1570	-	-	144.9	124.4
Δ (100 → 0%)	-0.0045	-0.0430	+0.0038	+0.0430	1.18	1.00	-	-
Δ (%)	-6.1%	-45.6%	+4.0%	+37.8%	-	-	-	-

benefit of bypass suppression relative to the additional hydraulic cost as the OR decreased from 100% to 0%. As shown in Table 5, η_{OR} was 1.18

for FC-40 and 1.00 for GC-5X. Although FC-40 showed a higher η_{OR} , the absolute thermal resistance reduction of GC-5X was 0.0430 K·W⁻¹,

which is 9.5 times larger than that of FC-40, $0.0045 \text{ K}\cdot\text{W}^{-1}$. This indicates that OR control provides a substantially greater thermal benefit for the high-viscosity GC-5X despite the higher hydraulic penalty.

To further evaluate the overall thermal-hydraulic balance at each OR, the thermal-hydraulic figure of merit ($FOM = 1 / (R_{th} \cdot W_{pump})$) was introduced. A higher FOM indicates greater cooling performance per unit pumping power. As shown in Table 5, FOM increased as the OR decreased for both fluids in the plate fin heat sink. For FC-40, FOM increased slightly from 141.4 to 144.9 K^{-1} while GC-5X showed a much larger increase from 93.1 to 124.4 K^{-1} . Although GC-5X required a higher pumping-power increase, its substantial thermal resistance reduction outweighed the hydraulic penalty. This confirms that bypass suppression improved the overall thermal-hydraulic balance of GC-5X more effectively than that of FC-40.

Fig. 10 visualizes the local velocity distributions of FC-40 and GC-5X across varying OR, providing a physical basis for the thermal-hydraulic trends. As shown in Fig. 10 (a), FC-40 maintains a relatively stable flow pattern regardless of structural constraints due to its low viscosity. At OR 100%, the internal fin region avoids severe stagnation despite partial bypass, and the change in local velocity remains moderate as OR decreases to 0%, given that FC-40 already achieves partial penetration under unconstrained conditions. In contrast, GC-5X exhibits a far more dramatic response. At OR 100%, internal velocity remains extremely low, confirming the severe bypass phenomenon consistent with the stagnation zones identified in Fig. 7. At OR 50%, increased velocity below the heat sink indicates a physical rise in the internal mass flow rate, driving the significant 15.4 K temperature reduction through the superior thermophysical properties of GC-5X. At OR 0%, all flow momentum is forcibly concentrated into the heating region, with local velocities surging to levels comparable to FC-40. This mitigation of bypass allows the favorable thermophysical properties of GC-5X to be more effectively utilized at the fin surfaces, resulting in a greater

thermal improvement than that observed for FC-40 under the OR 0% condition.

The temperature contours in Fig. 10 (b) provide direct visual confirmation of these outcomes. FC-40 maintains relatively uniform heat sink temperatures across all OR conditions, while GC-5X shows a striking contrast: at OR 100%, an intense high-temperature region spans nearly the entire heat sink interior, substantiating the stagnation-induced thermal deterioration. As OR decreases to 0%, this high-temperature zone progressively recedes toward the thermal uniformity of FC-40, confirming that structural confinement is the decisive mechanism for converting the thermophysical potential of high-viscosity fluids into cooling performance.

To quantitatively support the bypass behavior observed in Fig. 10, the bypass mass flow fraction was calculated from the CFD results. In this analysis, the bypass region was defined as the flow region excluding the heat sink passages and the region directed toward the heat source. This definition was used because coolant flowing through the heat sink passages or near the heat source can still contribute to heat removal from the heat-spreading base. The bypass mass flow rate and bypass fraction were defined as,

$$\dot{m}_{bypass} = \dot{m}_{total} - \dot{m}_{active}$$

$$\Phi_{bypass} = \left(\frac{\dot{m}_{bypass}}{\dot{m}_{total}} \right) \times 100$$

respectively. A lower Φ_{bypass} indicates greater coolant penetration into the heat source region.

As summarized in Table 6, under OR 100%, GC-5X exhibited a much higher bypass fraction of 89.69%, whereas FC-40 showed 48.41%. This result quantitatively confirms that most of the GC-5X flow bypassed the active heat-sink region under the open configuration. Therefore, although GC-5X has higher specific heat and thermal conductivity than

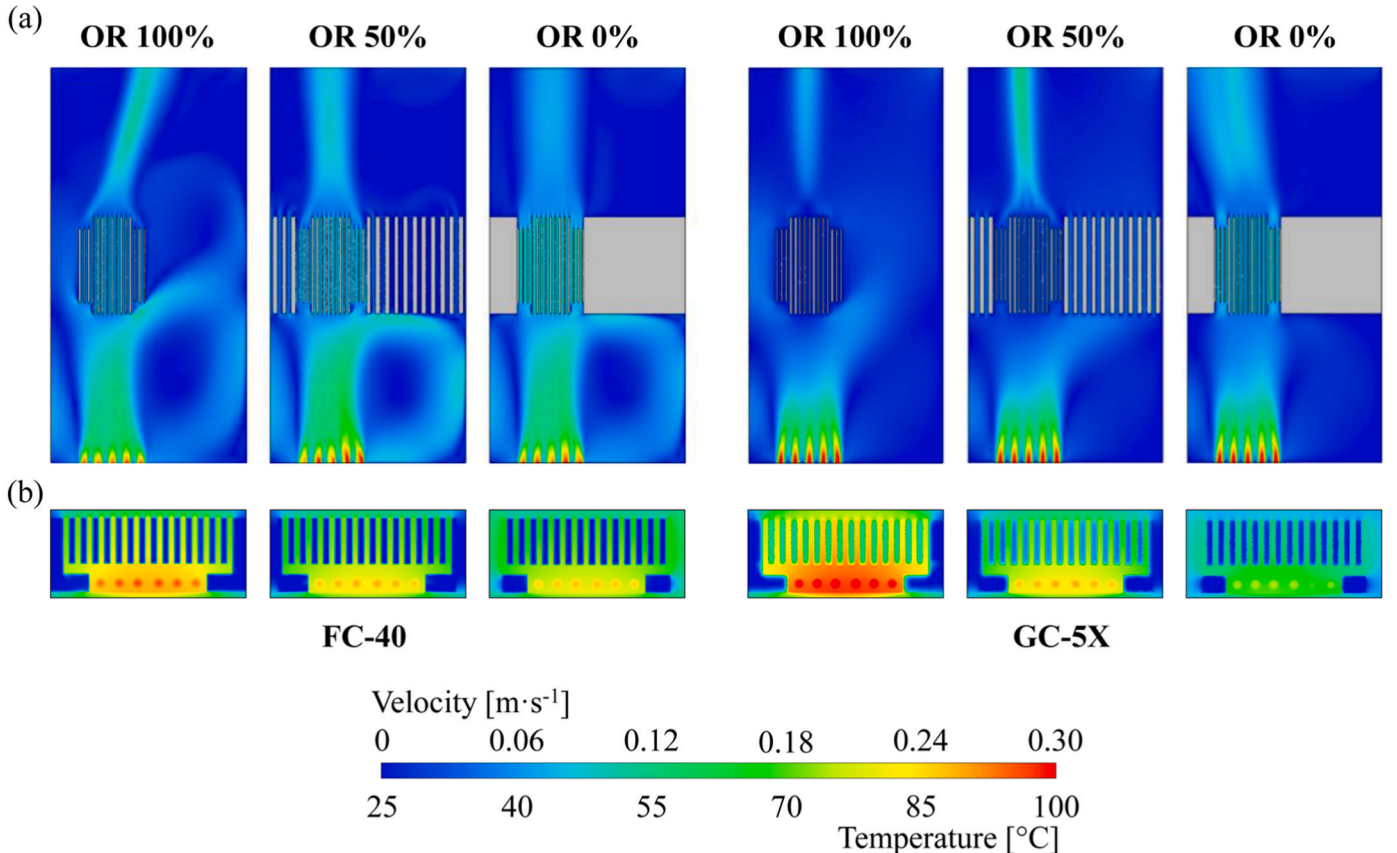


Fig. 10. (a) Velocity and (b) temperature contour across different open ratios for FC-40 and GC-5X.

Table 6

Bypass mass flow rate and bypass fraction as a function of open ratio in the plate fin heat sink.

Parameter	$\dot{m}_{\text{total}}$ (kg·s ⁻¹)		$\dot{m}_{\text{bypass}}$ (kg·s ⁻¹)		Φ_{bypass} (%)	
	FC-40	GC-5X	FC-40	GC-5X	FC-40	GC-5X
OR 100%	0.15459	0.06917	0.07483	0.06204	48.41	89.69
OR 50%			0.05666	0.04648	36.65	67.20
OR 0%			0.00120	0.00055	0.78	0.79

FC-40, these favorable properties were not effectively utilized at the heat-transfer surfaces because coolant penetration into the heat sink was insufficient. As the OR decreased, the bypass fraction was significantly reduced for both fluids. At OR 0%, Φ_{bypass} decreased to 0.78% for FC-40 and 0.79% for GC-5X, indicating that OR reduction effectively forced the coolant into the heat source region. Consequently, GC-5X exhibited a

45.6% reduction in thermal resistance, much larger than the 6.1% reduction observed for FC-40. These results quantitatively demonstrate that the thermal advantage of GC-5X is conditional on sufficient coolant penetration, rather than being guaranteed by thermophysical properties alone.

Fig. 11 presents the effect of volume flow rate on chip temperature, thermal resistance, and pressure drop across three OR conditions for FC-40 and GC-5X, providing insight into how structural bypass control interacts with flow rate to determine system performance.

For FC-40 in Fig. 11 (a), thermal performance improves consistently with increasing flow rate across all OR conditions, with the performance gap between OR conditions remaining relatively stable throughout the flow rate range. The chip temperature decreases from 77.4°C to 75.0°C at OR 100%, 77.0°C to 73.0°C at OR 50%, and 76.9°C to 70.4°C at OR 0% as flow rate increases from 6 to 14 LPM. This consistent improvement reflects the low viscosity of FC-40, which enables effective fin

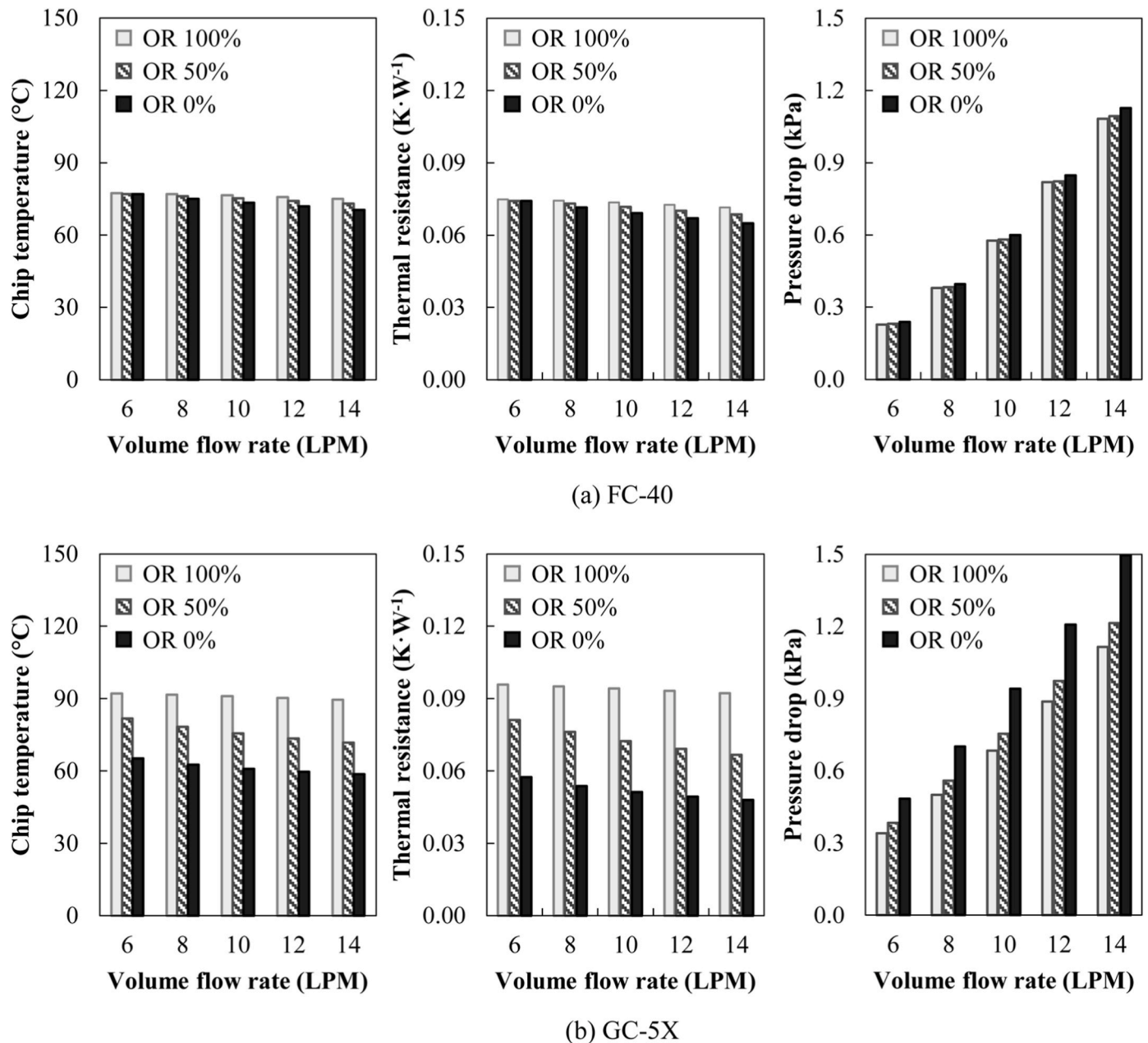


Fig. 11. Numerical results of volume flow rate effects on chip temperature, thermal resistance and pressure drop under different open ratio conditions for (a) FC-40 and (b) GC-5X.

Table 7

Thermal resistance and pumping power to open ratio under MF fin condition.

Parameter	R_{th} ($K \cdot W^{-1}$)		W_{pump} (W)		η_{OR} ($K \cdot W^{-1}/W$)		FOM (K^{-1})	
	FC-40	GC-5X	FC-40	GC-5X	FC-40	GC-5X	FC-40	GC-5X
OR 100%	0.0461	0.1247	0.0894	0.1141	-	-	242.6	70.3
OR 0%	0.0310	0.0490	0.1172	0.2048	-	-	275.5	99.6
Δ (100 $\rightarrow$ 0%)	-0.0151	-0.0757	+0.0279	+0.0908	0.54	0.83	-	-
Δ (%)	-32.8%	-60.7%	+31.2%	+79.6%	-	-	-	-
Plate fin (Table 5)								
Δ (100 $\rightarrow$ 0%)	-0.0045	-0.0430	+0.0038	+0.0430	1.18	1.00	-	-
Δ (%)	-6.1%	-45.6%	+4.0%	+37.8%	-	-	-	-

penetration regardless of OR condition, allowing increased flow rate to be directly converted into thermal gain. Hydraulically, pressure drop increases with flow rate across all conditions, with OR 0% reaching 1.13 kPa at 14 LPM.

GC-5X in Fig. 11 (b) exhibits markedly different behavior. At OR

100%, chip temperature remains largely insensitive to flow rate, decreasing only marginally from 92.1°C to 89.6°C — consistent with the bypass-dominated behavior. In contrast, OR 50% and OR 0% show progressively stronger flow-rate sensitivity, with chip temperatures decreasing from 81.8°C to 71.7°C and 65.2°C to 58.6°C, respectively.

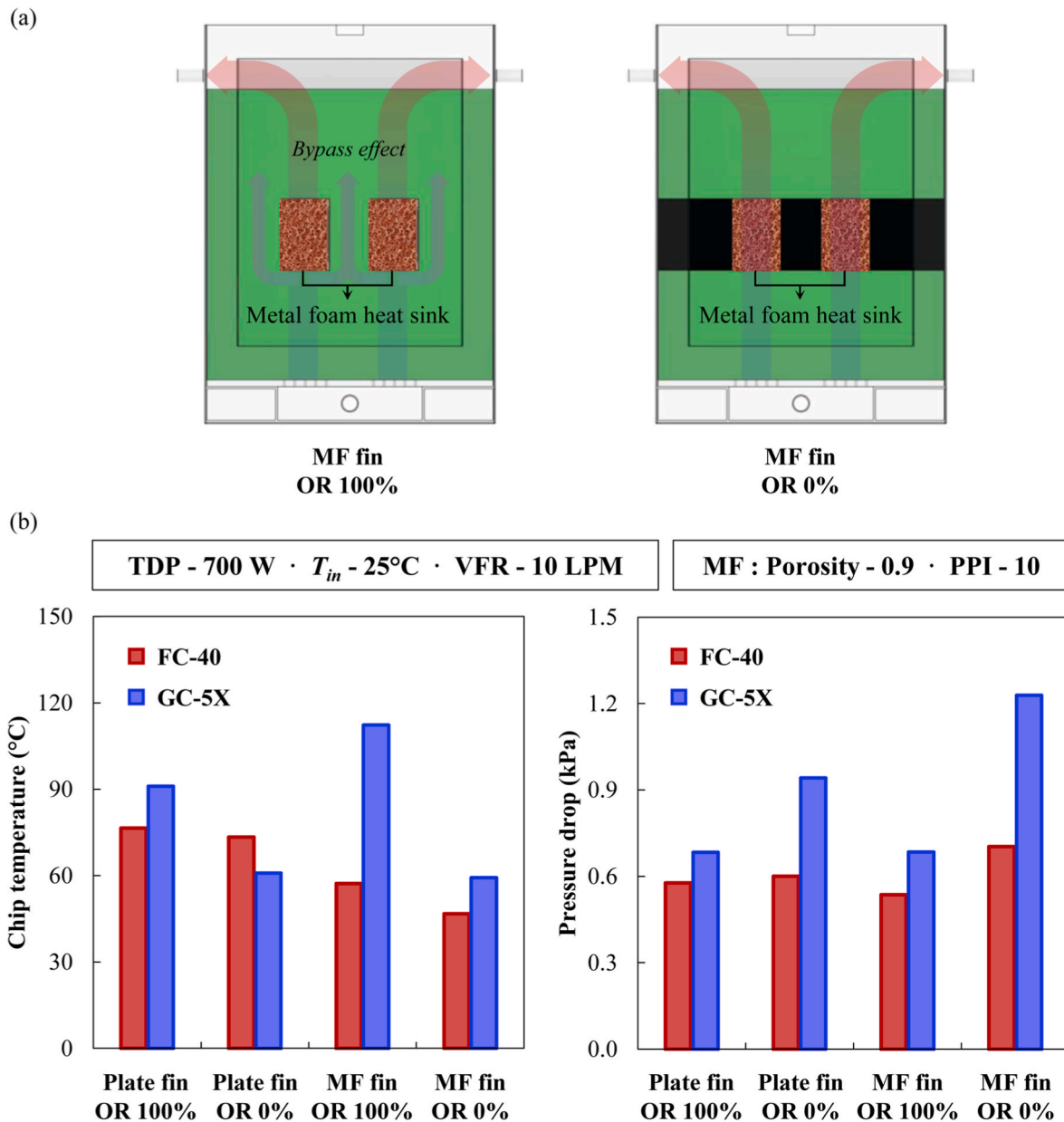


Fig. 12. (a) Schematic of metal foam assisted systems and (b) chip temperature and pressure drop under combined open ratio and heat sink structure conditions.

This confirms that structural confinement is a prerequisite for GC-5X to convert increased flow rate into meaningful thermal gain. Hydraulically, OR 0% incurs the highest penalty, with pressure drop reaching 1.50 kPa at 14 LPM, while OR 100% and OR 50% show lower and more comparable pressure profiles. These results demonstrate that while FC-40 benefits from flow rate augmentation under any OR condition, GC-5X requires sufficient structural confinement to activate its flow-rate sensitivity — further reinforcing that the optimal degree of bypass suppression is governed by fluid viscosity.

3.3. Metal foam application on heat sink

To explore the potential of metal foam as an alternative heat sink structure for further enhancing heat transfer performance, a metal foam-assisted configuration was numerically investigated alongside the plate fin systems, with a baseline foam structure of 10 PPI and a porosity of 0.9. Fig. 12 (a) illustrates the two representative configurations — MF fin at OR 100% and OR 0% — highlighting the role of the flow control plate in determining internal flow distribution within the porous medium.

Fig. 12 (b) presents the chip temperature and pressure drop across

the four combined conditions. For FC-40, metal foam integration yields a significant temperature reduction regardless of OR condition, achieving 57.3°C and 46.7°C at OR 100% and OR 0%, respectively, compared to 76.5°C and 73.3°C for the plate fin configurations. This suggests that the low viscosity of FC-40 enables effective foam penetration even without structural confinement, potentially allowing full utilization of the expanded heat transfer surface area. For GC-5X, the results reveal a strikingly contrasting behavior. At MF fin OR 100%, the chip temperature surges to 112.3°C — substantially higher than any plate fin condition — as high viscous resistance causes severe flow stagnation at the foam interface. However, at MF fin OR 0%, forced injection through the foam interstices produces a dramatic recovery to 59.3°C, suggesting that OR control may be essential rather than merely beneficial for high-viscosity fluids in metal foam configurations.

Table 7 quantifies these observations in terms of thermal resistance, pumping power, and thermal-hydraulic performance. For GC-5X in the MF fin configuration, OR reduction from 100% to 0% yielded a thermal resistance decrease of 0.0757 K·W⁻¹, corresponding to a 60.7% reduction. This reduction was 1.3 times larger than that of the plate fin counterpart, 45.6%. The OR efficiency index η_{OR} reached 0.83 K·W⁻¹/W for GC-5X in the MF fin configuration, compared to the plate fin value of

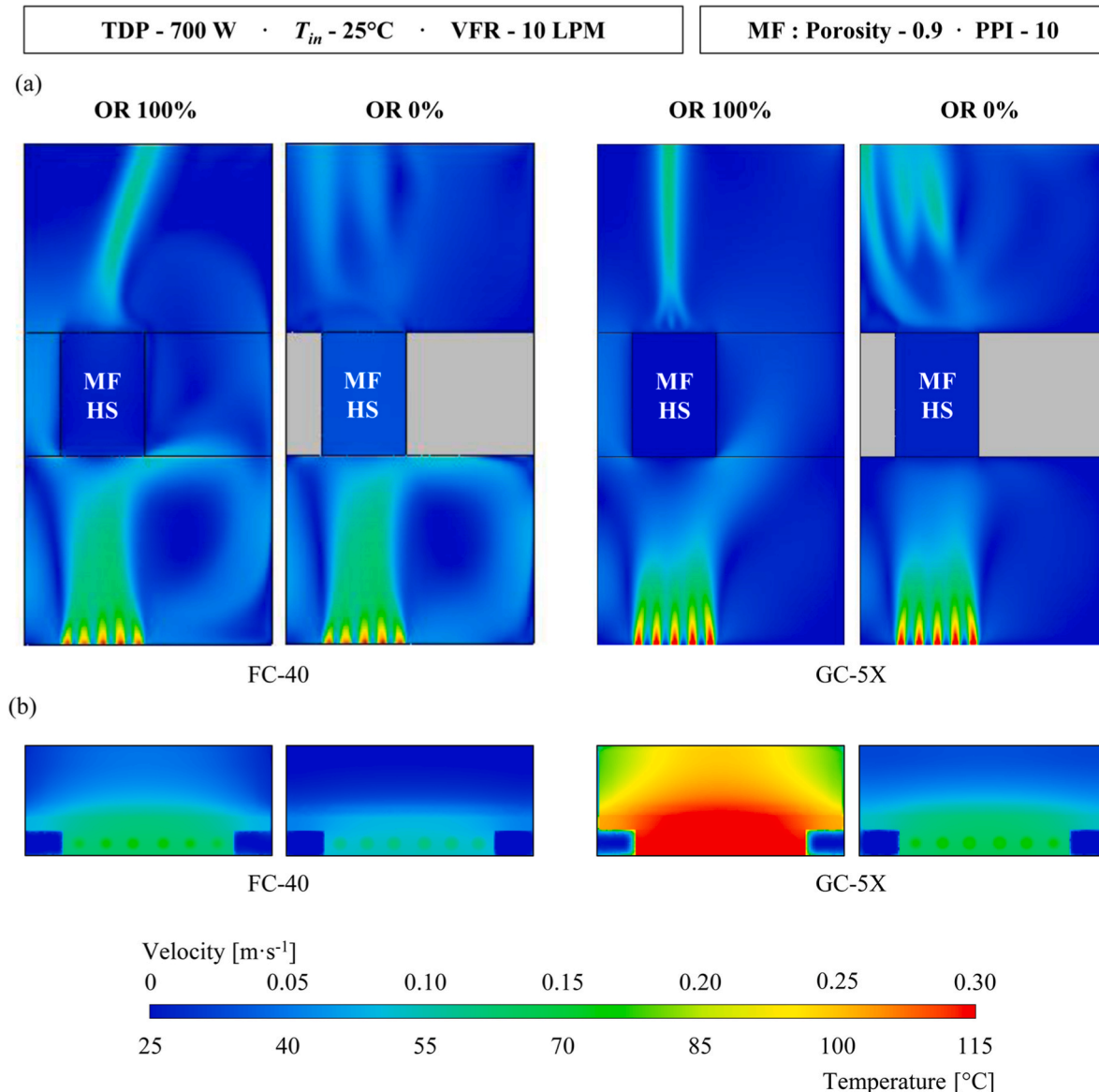


Fig. 13. (a) Velocity and (b) temperature contour for different open ratio in metal foam assisted system.

Table 8

Bypass mass flow rate and bypass fraction as a function of open ratio in the metal-foam heat sink.

Parameter	$\dot{m}_{\text{total}}$ (kg·s ⁻¹)		$\dot{m}_{\text{bypass}}$ (kg·s ⁻¹)		Φ_{bypass} (%)	
	FC-40	GC-5X	FC-40	GC-5X	FC-40	GC-5X
OR 100%	0.15459	0.06917	0.10072	0.06382	65.15	92.26
OR 0%			0.00932	0.00446	6.03	6.45

1.00 K·W⁻¹/W. For FC-40, the MF fin configuration yielded a more substantial thermal resistance reduction of 32.8% alongside a pumping power increase of 31.2%, resulting in an η_{OR} of 0.54 K·W⁻¹/W. Although both fluids exhibited comparable η_{OR} values in the MF fin configuration, the absolute thermal resistance reduction of GC-5X, 0.0757 K·W⁻¹, was 5.0 times larger than that of FC-40, 0.0151 K·W⁻¹, indicating that OR control still provides a substantially greater thermal benefit for the high-viscosity fluid despite the stronger hydraulic penalty.

The *FOM* further clarifies this trend. For GC-5X, *FOM* increased from 70.3 to 99.6 K⁻¹ after OR reduction, indicating that the thermal resistance reduction outweighed the pumping power penalty. For FC-40, the thermal resistance decreased by 32.8% while pumping power increased by 31.2%, resulting in a moderate *FOM* increase from 242.6 to 275.5 K⁻¹. Although both fluids showed improved *FOM* after OR reduction in the MF fin configuration, the relative gain was more pronounced for GC-5X in terms of absolute thermal resistance reduction, whereas FC-40 maintained a consistently higher *FOM* across all conditions owing to its lower viscosity enabling more effective foam penetration even at OR 100%. These results indicate that in metal foam configurations, OR suppression improves the thermal-hydraulic balance for both fluids, but the governing mechanism differs: GC-5X benefits primarily from the recovery of severely stagnated flow, while FC-40 benefits from incremental enhancement of already-active foam penetration.

The velocity and temperature contours in Fig. 13 provide a physical basis for the performance trends identified in Fig. 12. As shown in the velocity distributions in Fig. 13 (a), bypass phenomenon is observed at

OR 100% for both fluids, where a significant portion of the fluid avoids the high resistance of the metal foam and escapes through the side channels. For FC-40, internal flow within the foam remains at a moderate velocity even at OR 100%, indicating partial penetration into the porous structure. In contrast, GC-5X exhibits near-stagnant internal flow at OR 100%, failing to overcome the frontal resistance of the foam — a flow environment considerably more restrictive than that of the plate fin configuration. At OR 0%, forced injection through the foam interstices produces a drastic increase in localized velocity for both fluids, effectively eliminating the stagnation regions previously observed in GC-5X.

The temperature contours in Fig. 13 (b) directly reflect these flow characteristics. At OR 100%, FC-40 maintains a relatively moderate temperature distribution within the foam, whereas GC-5X exhibits a severely elevated temperature field spanning nearly the entire foam region, consistent with the 112.3°C chip temperature. At OR 0%, both fluids show markedly suppressed temperature distributions, with GC-5X demonstrating a dramatic recovery that aligns with its 59.3°C chip temperature. These contours confirm that the extreme thermal deterioration of GC-5X at OR 100% and its subsequent recovery at OR 0% are directly attributable to viscosity-driven flow stagnation and its structural resolution, reinforcing the conclusion that OR control is a critical design parameter for high-viscosity fluids in metal foam-assisted configurations. These findings suggest that OR sensitivity within the MF-assisted system is fundamentally governed by fluid viscosity — a relationship that is further quantified in terms of thermal resistance and pumping power trade-offs.

The bypass fraction for the metal foam heat sink is summarized in Table 8. Compared with the plate fin configuration, the metal foam heat sink showed a stronger dependence on coolant penetration because the porous structure introduced additional flow resistance. Under OR 100%, the bypass fraction reached 92.26% for GC-5X and 65.15% for FC-40, indicating that most of the high-viscosity GC-5X bypassed the active metal-foam cooling region when bypass paths were available. This severe bypass flow prevented GC-5X from utilizing the extended heat-transfer area of the metal foam, resulting in a chip temperature of

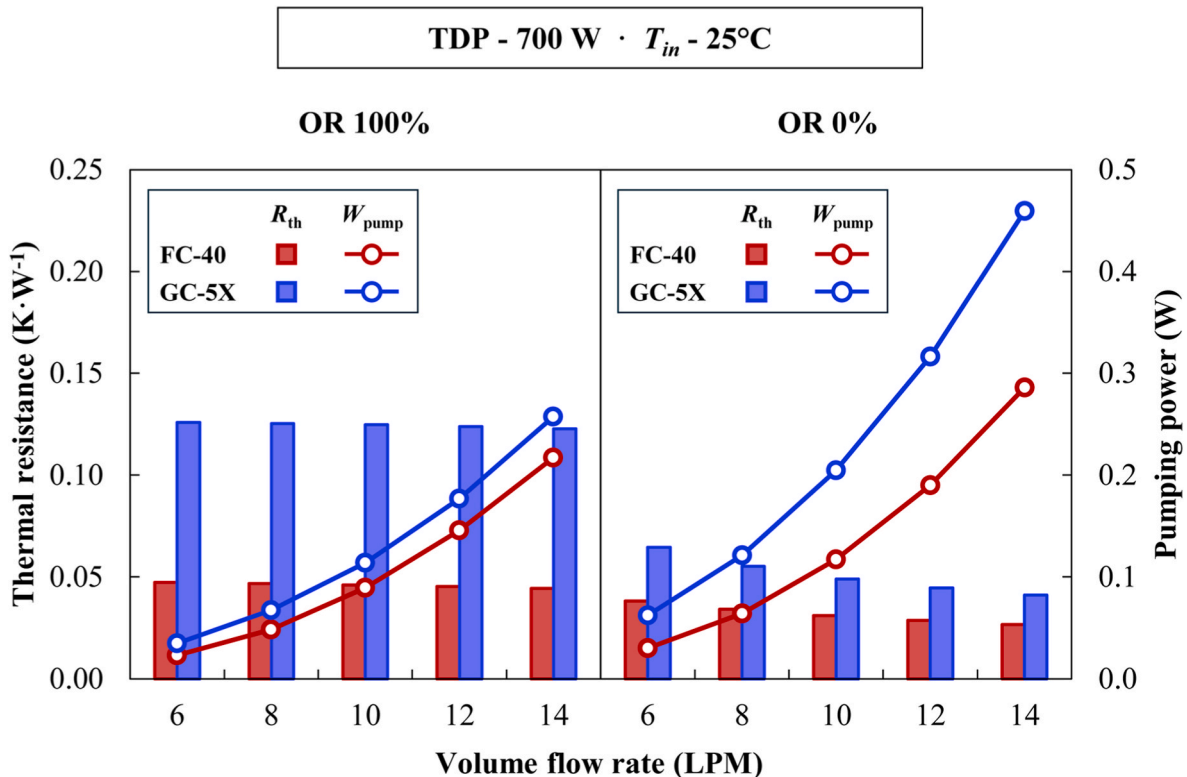


Fig. 14. Effect of volume flow rate on thermal resistance and pumping power in metal foam-assisted immersion cooling under different open ratio conditions.

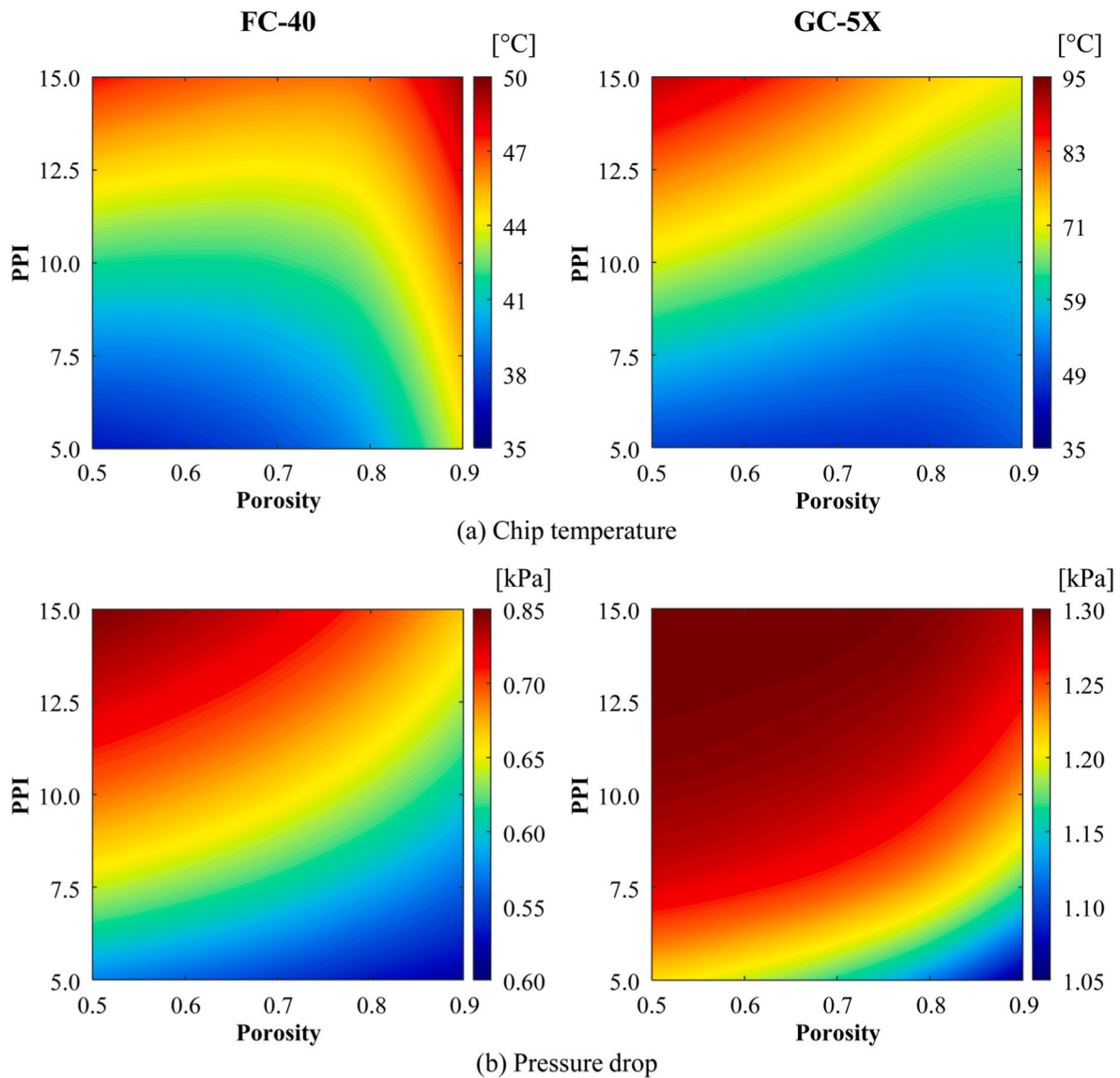


Fig. 15. Influence of metal foam porosity and PPI on (a) chip temperature and (b) pressure drop.

112.3°C despite its higher specific heat and thermal conductivity. After OR reduction to 0%, the bypass fraction sharply decreased to 6.45% for GC-5X and 6.03% for FC-40. Correspondingly, the chip temperature of GC-5X recovered to 59.3°C. These results demonstrate that the poor performance of GC-5X with metal foam under OR 100% was not caused by inferior heat-storage or conduction properties, but by severe bypass flow and underutilization of the metal-foam structure. Therefore, sufficient coolant penetration is essential for realizing the thermal benefits of dielectric fluids with superior thermal properties in metal foam assisted system.

Fig. 14 presents the thermal resistance and pumping power as a function of volume flow rate for the metal foam-assisted configuration under OR 100% and OR 0% conditions. At OR 100%, both fluids exhibit nearly invariant thermal resistance across the full flow rate range — decreasing by only 6.0% for FC-40 and 2.5% for GC-5X from 6 to 14 LPM — while pumping power rises sharply, mirroring the bypass-dominated behavior observed in the open-tank configuration. This confirms that the high flow resistance of the metal foam further amplifies the bypass phenomenon, rendering flow rate augmentation ineffective when structural confinement is absent.

At OR 0%, the behavior shifts markedly. Thermal resistance decreases substantially with increasing flow rate — by 30.1% for FC-40

and 36.2% for GC-5X — demonstrating that forced injection through the foam structures enables the additional flow to contribute directly to heat removal. Notably, GC-5X exhibits greater thermal resistance sensitivity to flow rate than FC-40 under full confinement, suggesting that the thermophysical advantages of the high-viscosity fluid are more effectively mobilized as inertial forces increase. However, the pumping power of GC-5X rises more steeply than that of FC-40 at OR 0%, reaching 0.459 W at 14 LPM compared to 0.286 W, reflecting the disproportionate hydraulic cost imposed by high viscosity under full structural confinement. These results collectively reinforce that OR control is a prerequisite for effective flow rate utilization in metal foam-assisted systems, and that the trade-off between thermal gain and hydraulic penalty is fundamentally governed by fluid viscosity.

To investigate the influence of metal foam structures on cooling and pressure drop performance, the effects of varying PPI and porosity were systematically analyzed within a range of 5 to 15 PPI and 0.5 to 0.9 porosity. These structural parameters directly dictate flow resistance, which fundamentally governs the bypass phenomenon where coolant fails to penetrate the foam and escapes through the sides.

Analysis of the chip temperature contours in Fig. 15 (a) reveals that FC-40 and GC-5X exhibit opposing thermal responses. For FC-40, the minimum temperature is observed in moderately dense regions where

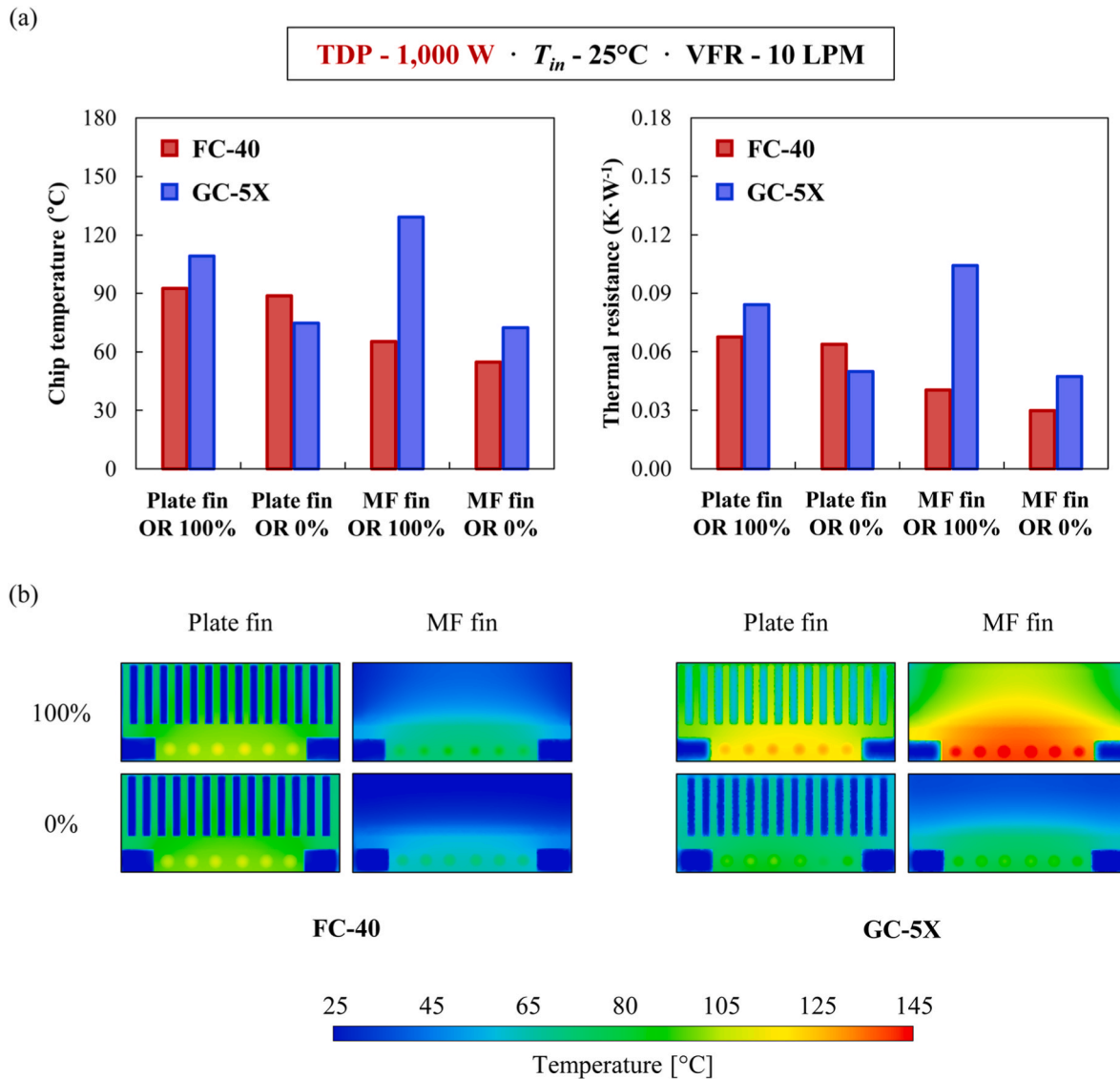


Fig. 16. Thermal performance under the high-power-density condition: (a) chip temperature and thermal resistance and (b) temperature contours.

surface area is sufficient, but flow resistance remains low enough for penetration. If PPI becomes excessively high, flow resistance exceeds a critical threshold, reducing the actual mass flow rate entering the heat sink and degrading cooling performance despite the expanded surface area. In contrast, GC-5X reacts much more sensitively to flow resistance. In low-porosity, high-PPI regions, fluid flow stagnates severely, causing chip temperature to surge to 91.1°C. As a result, its thermal optimum is shifted toward sparse structures that prioritize flow penetration over surface area maximization — consistent with the viscosity-driven bypass sensitivity established throughout this section.

Fig. 15 (b) confirms the hydraulic cost associated with denser foam structures, showing a sharp pressure drop increase as the metal foam becomes denser. While GC-5X incurs higher absolute hydraulic costs, its best cooling performance is achieved in lower pressure-drop regions where flow entry is secured. These results establish that the optimal foam structure must be tailored to the specific viscosity of the working fluid: FC-40 benefits from moderately dense configurations, whereas GC-5X strictly requires sparse designs to prevent flow bypass. These results highlight the need for fluid-specific metal foam design, as the optimal porous structure depends strongly on the flow behavior and penetration characteristics of each dielectric fluid.

3.4. High-power density condition

To evaluate the feasibility of cooling next-generation high-performance chips, the thermal performance of each configuration was analyzed under a high-power density condition of 1000 W TDP at a fixed volumetric flow rate of 10 LPM. Fig. 16 (a) presents the chip temperature and thermal resistance across the four structural conditions for both fluids, with the corresponding temperature contours shown in Fig. 16 (b).

For FC-40, the Plate fin OR 100% and OR 0% configurations fail to provide adequate cooling, with chip temperatures reaching 92.6°C and 88.8°C, respectively. The MF fin OR 100% condition reduces this to 65.4°C, and the MF fin OR 0% configuration achieves the lowest chip temperature of 54.8°C with a thermal resistance of 0.0298 K·W⁻¹, attributable to FC-40's low viscosity enabling deep foam penetration under full confinement. The temperature contours confirm a progressively uniform and suppressed thermal distribution as structural confinement increases, with MF fin OR 0% achieving the most favorable heat distribution across the foam matrix.

For GC-5X, the performance gap between configurations is considerably more pronounced. The plate fin OR 100% condition records 109.2°C, while OR 0% reduces this to 74.8°C. The MF fin OR 100%

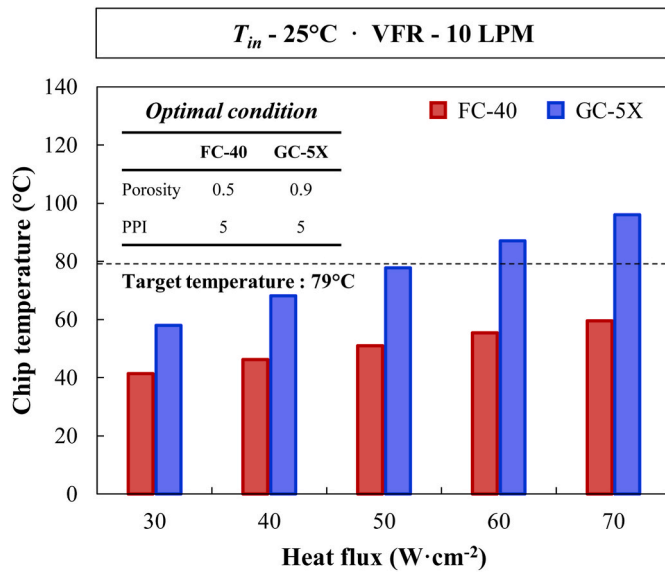


Fig. 17. Thermal performance of MF fin OR 0% configuration under varying heat flux conditions.

condition produces a severely elevated temperature of 129.3°C — the worst of all configurations — as the combined effect of high viscosity and foam resistance creates an extreme flow barrier, visually confirmed by the intense thermal concentration in the contours. In contrast, MF fin OR 0% achieves 72.4°C with a thermal resistance of 0.0474 K·W⁻¹, a reduction of 43.7% relative to the plate fin OR 100% baseline, with the contours showing a dramatic suppression of high-temperature zones approaching the thermal uniformity of FC-40. These numerical results suggest that under kilowatt-scale heat loads, the combination of metal foam and full bypass suppression represents the most promising pathway toward effective thermal management.

It was confirmed that the MF fin OR 0% system exhibits the most stable cooling performance under high heat flux conditions. Accordingly, the optimal metal foam structure was implemented for each dielectric fluid, and the average chip temperature as a function of heat flux was calculated. Based on the results in Fig. 15, a porosity of 0.5 and 5 PPI were applied for FC-40, while a porosity of 0.9 and 5 PPI were utilized for GC-5X.

As illustrated in Fig. 17, both fluids successfully manage high heat flux conditions, albeit with distinct thermal characteristics. For FC-40, the average chip temperature remains remarkably stable, ranging from approximately 41.4°C at 30 W·cm⁻² to 59.6°C at 70 W·cm⁻², maintaining a significant thermal margin of approximately 19.4°C below the 79°C reliability threshold at maximum load [22]. In contrast, GC-5X exhibits a steeper temperature rise with increasing heat flux, maintaining 58.1°C at 30 W·cm⁻² but reaching 77.8°C at 50 W·cm⁻², approaching the 79°C reliability threshold, and further escalating to

96.0°C at 70 W·cm⁻².

The performance discrepancy stems from the synergy between thermophysical properties and foam architecture. FC-40's lower viscosity enables effective utilization of the dense 0.5 porosity structure, maximizing surface area and turbulent mixing. Conversely, GC-5X requires the sparser 0.9 porosity configuration to maintain adequate permeability, resulting in lower effective surface area and reduced mixing intensity. These results suggest that viscosity-structure compatibility is the decisive factor governing the practical heat flux ceiling, with FC-40 demonstrating stable coverage up to 70 W·cm⁻² while GC-5X approaches its practical limit at approximately 50 W·cm⁻², with the threshold margin nearly exhausted beyond this point.

4. Conclusions

This study presents the first systematic investigation of single-phase immersion cooling under chip-level heat loads exceeding 700 W per chip, introducing the open ratio as the first quantitative metric for bypass control and extending the analysis to metal foam-assisted configurations across FC-40 and GC-5X dielectric fluids.

Table 9 summarizes the thermal and hydraulic performance of both fluids across all structural conditions at baseline operating conditions (TDP: 700 W, T_{in}: 25°C, VFR: 10 LPM). While low-viscosity FC-40 offers stable performance across all configurations, the potential of high-viscosity GC-5X is only realized when bypass flow is suppressed through structural confinement as heat flux demands escalate toward kilowatt-scale levels — a finding that motivates the design framework established in this study.

The following significant findings were obtained:

1. In the 3U-scale open configuration, a significant flow bypass effect was observed as the coolant followed paths of least resistance. The GC-5X recorded 99.6°C at 700 W TDP — 23.0 K higher than FC-40 — despite superior thermophysical properties. Flow rate augmentation from 11 to 19 LPM raised pumping power from 0.17859 to 0.69159 W while thermal resistance remained nearly invariant at approximately 0.106 K·W⁻¹.
2. The OR sweep from 100% to 0% revealed thermal resistance reductions of 45.6% for GC-5X and 6.1% for FC-40 — a 7.5-fold difference — with pumping power increases of 37.8% and 4.0%, respectively. At OR 0%, performance inversion occurred with GC-5X achieving 60.9°C versus 73.3°C for FC-40. Under full confinement, GC-5X chip temperature decreased from 65.2°C to 58.6°C across 6 to 14 LPM, confirming that OR control activates flow-rate sensitivity otherwise absent at OR 100%.
3. In the metal foam configuration, OR sensitivity was amplified: GC-5X reached 112.3°C at OR 100% — worse than any plate fin condition — yet recovered to 59.3°C at OR 0%, with an absolute thermal resistance reduction 5.0 times larger than that of FC-40. Optimal foam structure was confirmed to be viscosity-dependent, with FC-40 favoring low porosity and GC-5X requiring high porosity.
4. Under 1000 W TDP, MF fin OR 0% was the only configuration maintaining stable operation for both fluids, achieving 54.8°C for FC-40 and 72.4°C for GC-5X — a 43.7% thermal resistance reduction relative to the plate fin OR 100% baseline. FC-40 sustained a 19.4°C margin below the 79°C reliability threshold up to 70 W·cm⁻², while GC-5X approached the threshold at 50 W·cm⁻², with the margin nearly exhausted beyond this point.
5. Overall, the thermal effectiveness of single-phase immersion cooling under high chip-level heat loads was governed by the coupled effect of dielectric fluid properties, coolant penetration, and structural flow control, rather than by thermophysical properties alone. Although GC-5X possesses higher specific heat and thermal conductivity than FC-40, its thermal advantage was realized only when bypass flow was sufficiently suppressed and coolant penetration into the heat sink or metal foam was secured.

Table 9

Performance summary of FC-40 and GC-5X under combined heat sink and open ratio conditions.

Dielectric	Heat sink design	Open ratio	Main output			
			T(°C)	R _{th} (K·W ⁻¹)	W _{pump} (W)	FOM (K ⁻¹)
FC-40	Plate fin	100%	76.5	0.0736	0.0961	141.4
		0%	73.3	0.0691	0.1000	144.9
	MF fin	100%	57.3	0.0461	0.0894	242.6
		0%	46.7	0.0310	0.1172	275.5
GC-5X	Plate fin	100%	91.0	0.0942	0.1139	93.1
		0%	60.9	0.0512	0.1570	124.4
	MF fin	100%	112.3	0.1247	0.1141	70.3
		0%	59.3	0.0490	0.2048	99.6

Future work will extend the immersion cooling framework established in this study across several directions. First, experimental validation of the metal foam-assisted configuration remains the most immediate priority, as the present results are based entirely on numerical predictions. Second, system-level energy metrics including PUE and rack-level pumping efficiency should be incorporated to evaluate the total energy impact in real data center environments. Finally, FC-40 is subject to increasingly stringent environmental regulations due to its high global warming potential, limiting its long-term viability as a coolant. The viscosity-structure compatibility framework established in this study provides a rational basis for identifying or developing oil-based dielectric fluids with higher density and lower viscosity that could leverage the structural confinement strategies demonstrated here to reconcile thermal performance with environmental compliance.

CRediT authorship contribution statement

Inwoo Chun: Writing – original draft, Visualization, Validation,

Software, Methodology, Investigation, Formal analysis. **Hongseok Choi:** Methodology, Investigation. **Yongjoo Jun:** Validation, Investigation. **Sangwook Lee:** Software, Investigation. **Hoseong Lee:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Funding

This work was supported by the Korea Institute of Energy Technology Evaluation and Planning (KETEP) and the Ministry of Trade, Industry & Energy (MOTIE) of the Republic of Korea (No. RS-2024-00397362, No. RS-2026-25536204).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

APPENDIX A. Dielectric fluid properties

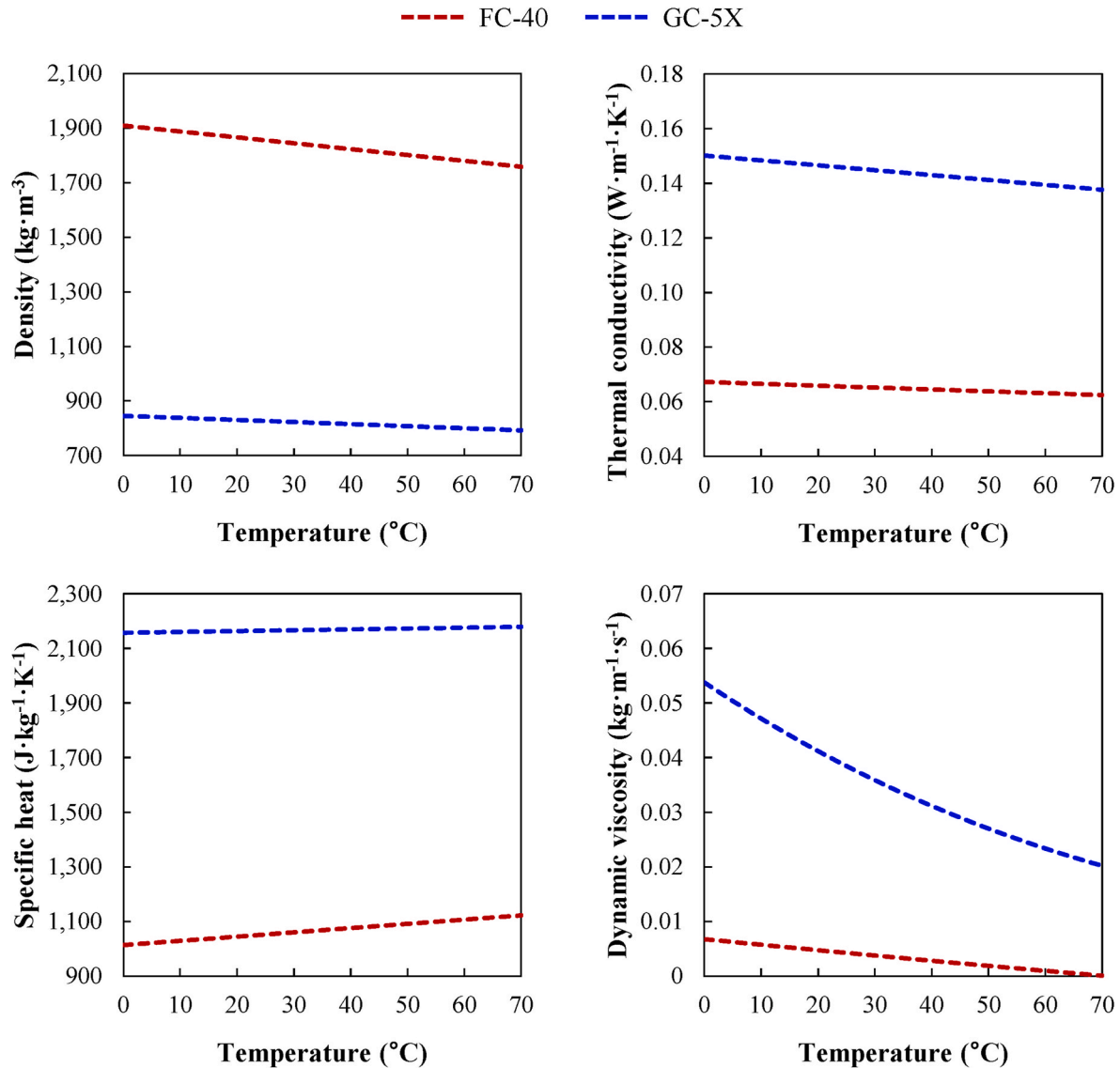


Fig. A. 1. Thermophysical properties of dielectric fluids.

Data availability

Data will be made available on request.

References

- [1] Wang N, Wei Z, Tian B, Dong Z, Shao S. Investigation on multi-objective optimization of thermal-hydraulic performance for enhanced heat transfer in jet impingement cooling. *Energy Convers Manag* 2025;340:119999. <https://doi.org/10.1016/j.enconman.2025.119999>.
- [2] Zhang B, Li H, Xu T, Wang L, Chen L, Li Z. Integrated CFD modeling and experimental validation of single-phase immersion cooling for data center thermal management. *Energy* 2026;347:140315. <https://doi.org/10.1016/j.energy.2026.140315>.
- [3] Deng Z, Zhang S, Ma K, Jia CB, Sun Y, Chen X, et al. Numerical and experimental study on cooling high power chips of data centers using double-side cooling module based on mini-channel heat sink. *Appl Therm Eng* 2023;227:120282. <https://doi.org/10.1016/j.applthermaleng.2023.120282>.
- [4] Haghshenas K, Taheri S, Goudarzi M, Mohammadi S. Infrastructure aware heterogeneous-workloads scheduling for data center energy cost minimization. *IEEE Trans Cloud Comput* 2022;10:972–83. <https://doi.org/10.1109/TCC.2020.2977040>.
- [5] Koot M, Wijnhoven F. Usage impact on data center electricity needs: a system dynamic forecasting model. *Appl Energy* 2021;291:116798. <https://doi.org/10.1016/j.apenergy.2021.116798>.
- [6] Huang Y, Ge J, Chen Y, Zhang C. Natural and forced convection heat transfer characteristics of single-phase immersion cooling systems for data centers. *Int J Heat Mass Tran* 2023;207. <https://doi.org/10.1016/j.ijheatmasstransfer.2023.124023>.
- [7] Ruichek A, Hnayno M, Chehade A, Fardoun F, Moussa T, Polidori G, et al. Immersion cooling fluid comparison using a single-phase immersion/liquid technique for data centers. *Appl Therm Eng* 2025;280:128355. <https://doi.org/10.1016/j.applthermaleng.2025.128355>.
- [8] Li J, Luo X, Wang M, Chen C, Yang C, Zhang Z, et al. Study of flow and heat transfer characteristics of tandem cold plates for data center cooling. *Case Stud Therm Eng* 2025;73:106590. <https://doi.org/10.1016/j.csite.2025.106590>.
- [9] Heydari A, Gharaibeh AR, Tradat M, soud Q, Manaserh Y, Radmard V, et al. Experimental evaluation of direct-to-chip cold plate liquid cooling for high-heat-density data centers. *Appl Therm Eng* 2024;239:122122. <https://doi.org/10.1016/j.applthermaleng.2023.122122>.
- [10] Tian T, Gui C, Wang Y, Shen C, Wang R, Wu T, et al. Investigation on negative pressure cold plate liquid cooling solution for data center application. 2025 24th IEEE intersoc conf Therm thermomechanical Phenom Electron syst. 2025. p. 1–8. <https://doi.org/10.1109/ITHERM55376.2025.11235794>.
- [11] Habibi Khalaj A, Halmuge SK. A review on efficient thermal management of air- and liquid-cooled data centers: from chip to the cooling system. *Appl Energy* 2017; 205:1165–88. <https://doi.org/10.1016/j.apenergy.2017.08.037>.
- [12] Kanbur BB, Wu C, Fan S, Tong W, Duan F. Two-phase liquid-immersion data center cooling system: experimental performance and thermoeconomic analysis. *Int J Refrig* 2020;118:290–301. <https://doi.org/10.1016/j.ijrefrig.2020.05.026>.
- [13] Liu S, Guo S, Sun H, Xu Z, Li X, Wang X. Evaluation of energy, economic, and pollution emission for single-phase immersion cooling data center with different economizers. *Appl Therm Eng* 2025;260:125049. <https://doi.org/10.1016/j.applthermaleng.2024.125049>.
- [14] Zheng S, Su C, Yang X, Zhang Y, Duan K, Zhang Y, et al. A comprehensive review of single-phase immersion cooling in data centres. *Appl Therm Eng* 2025;272: 126385. <https://doi.org/10.1016/j.applthermaleng.2025.126385>.
- [15] Choi H, Kim H, Park P, Lee H. Development of a mini-channel and metal foam-assisted static immersion cooling and preheating system for electric vehicle battery thermal management. *Case Stud Therm Eng* 2025;70:106145. <https://doi.org/10.1016/j.csite.2025.106145>.
- [16] Choi H, Lee H, Kim J, Lee H. Hybrid single-phase immersion cooling structure for battery thermal management under fast-charging conditions. *Energy Convers Manag* 2023;287:117053. <https://doi.org/10.1016/j.enconman.2023.117053>.
- [17] Li X, Guo S, Sun H, Yin S, Liu S, Wang Z. Experimental study of the performance of liquid cooling tank used for single-phase immersion cooling data center. *Case Stud Therm Eng* 2024;63:105386. <https://doi.org/10.1016/j.csite.2024.105386>.
- [18] Huang Y, Liu B, Xu S, Bao C, Zhong Y, Zhang C. Experimental study on the immersion liquid cooling performance of high-power data center servers. *Energy* 2024;297:131195. <https://doi.org/10.1016/j.energy.2024.131195>.
- [19] Wang J, Xu S, Fan L, Gao M, Zhang W, He Y, et al. Thermal performance characteristic of single-phase immersion cooling on servers with optimized heatsink design. *Intersoc Conf Therm Thermomechanical Phenom Electron Syst ITherm* 2024:1–7. <https://doi.org/10.1109/ITHERM55375.2024.10709588>.
- [20] Zhou G, Zhou J, Huai X, Zhou F, Jiang Y. A two-phase liquid immersion cooling strategy utilizing vapor chamber heat spreader for data center servers. *Appl Therm Eng* 2022;210:118289. <https://doi.org/10.1016/j.applthermaleng.2022.118289>.
- [21] Wu X, Yang J, Liu Y, Zhuang Y, Luo S, Yan Y, et al. Investigations on heat dissipation performance and overall characteristics of two-phase liquid immersion cooling systems for data center. *Int J Heat Mass Tran* 2025;239:126575. <https://doi.org/10.1016/j.ijheatmasstransfer.2024.126575>.
- [22] Kim J, Choi H, Lee S, Lee H. Computational study of single-phase immersion cooling for high-energy density server rack for data centers. *Appl Therm Eng* 2025; 264:125476. <https://doi.org/10.1016/j.applthermaleng.2025.125476>.
- [23] Wang H, Yuan X, Zhang K, Lang X, Chen H, Yu H, et al. Performance evaluation and optimization of data center servers using single-phase immersion cooling. *Int J Heat Mass Tran* 2024;221:125057. <https://doi.org/10.1016/j.ijheatmasstransfer.2023.125057>.
- [24] Li X, Guo S, Sun H, Liu S, Wang X, Wang L. Comprehensive assessment of coolants used single-phase immersion cooling data center. *Energy* 2025;340:139240. <https://doi.org/10.1016/j.energy.2025.139240>.
- [25] Cheng CC, Chang PC, Li HC, Hsu FI. Design of a single-phase immersion cooling system through experimental and numerical analysis. *Int J Heat Mass Tran* 2020; 160:120203. <https://doi.org/10.1016/j.ijheatmasstransfer.2020.120203>.
- [26] Gandhi D, Chowdhury U, Chauhan T, Bansode P, Saini S, Shah JM, et al. Computational analysis for thermal optimization of server for single phase immersion cooling. *ASME 2019 int tech conf exhib packag Integr Electron Photonic microsystems, InterPACK* 2019. 2019. <https://doi.org/10.1115/InterPACK2019-6587>.
- [27] Ding B, Xia F, He L, Huang Y. A novel concept for performance enhancement of immersion-cooled data center servers. *Int Commun Heat Mass Tran* 2024;159: 107980. <https://doi.org/10.1016/j.icheatmasstransfer.2024.107980>.
- [28] Li X, Xu Z, Liu S, Zhang X, Sun H, Zhang X, et al. Optimization for single-phase immersion cooling data center. *Appl Therm Eng* 2023;224:120080. <https://doi.org/10.1016/j.applthermaleng.2023.120080>.
- [29] Liu S, Xu Z, Wang Z, Li X, Sun H, Zhang X, et al. Optimization and comprehensive evaluation of liquid cooling tank for single-phase immersion cooling data center. *Appl Therm Eng* 2024;245:122864. <https://doi.org/10.1016/j.applthermaleng.2024.122864>.
- [30] Ge J, Xia F, Zhang C, Huang Y. Performance enhancement of single-phase immersion liquid-cooled data center servers. *J Therm Sci* 2024;33:1757–72. <https://doi.org/10.1007/s11630-024-2010-4>.
- [31] Muneeswaran M, Lin YC, Wang CC. Performance analysis of single-phase immersion cooling system of data center using FC-40 dielectric fluid. *Int Commun Heat Mass Tran* 2023;145:106843. <https://doi.org/10.1016/j.icheatmasstransfer.2023.106843>.
- [32] Shrigondekar H, Lin YC, Wang CC. Investigations on performance of single-phase immersion cooling system. *Int J Heat Mass Tran* 2023;206:123961. <https://doi.org/10.1016/j.ijheatmasstransfer.2023.123961>.
- [33] Choi H, Jun Y, Chun H, Lee H. Comprehensive feasibility study on metal foam use in single-phase immersion cooling for battery thermal management system. *Appl Energy* 2024;375:124083. <https://doi.org/10.1016/j.apenergy.2024.124083>.
- [34] Jadhav PH, Gnanasekaran N, Mobedi M. Analysis of functionally graded metal foams for the accomplishment of heat transfer enhancement under partially filled condition in a heat exchanger. *Energy* 2023;263:125691. <https://doi.org/10.1016/j.energy.2022.125691>.
- [35] Tikadar A, Kumar S. Local hotspot thermal management using metal foam integrated heat sink. *Appl Therm Eng* 2023;221:119632. <https://doi.org/10.1016/j.applthermaleng.2022.119632>.
- [36] Saeedan M, Afshari E, Ziaei-Rad M. Modeling and optimization of turbulent flow through PEM fuel cell cooling channels filled with metal foam- a comparison of water and air cooling systems. *Energy Convers Manag* 2022;258:115486. <https://doi.org/10.1016/j.enconman.2022.115486>.
- [37] Ni D, Yu F. CFD simulation study of flow equalisation plate model in single-phase immersion liquid cooling for servers. *Therm Sci Eng Prog* 2024;47:102268. <https://doi.org/10.1016/j.tsep.2023.102268>.
- [38] Goodfellow. Material properties for indium foil. 2025.
- [39] MatWeb. Material property data — 304 stainless steel. 2025. <https://www.matweb.com/search/DataSheet.aspx?MatGUID=abc4415b0f8b490387e3c922237098da>.
- [40] Norouzi N. Assessment of technological path of hydrogen energy industry development: a review. *Iran J Energy Environ* 2021;12:273–84. <https://doi.org/10.5829/ijee.2021.12.04.01>.
- [41] enmove SK. GC-5X. 2025.
- [42] Lee S, Choi H, Park P, Kang S, Choi K, Lee H. Investigation of enhanced PCM heat sink design under extreme thermal conditions for high-heat-generating electronic devices. *Int Commun Heat Mass Tran* 2024;159:108050. <https://doi.org/10.1016/j.icheatmasstransfer.2024.108050>.
- [43] Pati S, Borah A, Boruah MP, Randive PR. Critical review on local thermal equilibrium and local thermal non-equilibrium approaches for the analysis of forced convective flow through porous media. *Int Commun Heat Mass Tran* 2022; 132:105889. <https://doi.org/10.1016/j.icheatmasstransfer.2022.105889>.